Chapter 9

Regression effect process

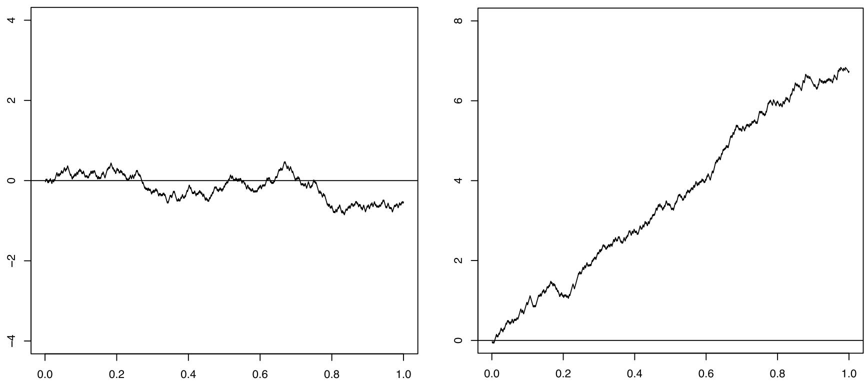
9.1 Chapter summary

In this chapter we describe the regression effect process. This can be established in different ways and provides all of the essential information that we need in order to gain an impression of departures from some null structure, the most common null structure corresponding to an absence of regression effect. Departures in specific directions enable us to make inferences on model assumptions and can suggest, of themselves, richer more plausible models. The regression effect process, in its basic form, is much like a scatterplot for linear regression telling us, before any formal statistical analysis, whether the dependent variable really does seem to depend on the explanatory variable as well as the nature, linear or more complex, of that relationship. Our setting is semi-parametric and the information on the time variable is summarized by its rank within the time observations. We make use of a particular time transformation and see that a great body of known theory becomes available to us immediately. An important objective is to glean what we can from a graphical presentation of the regression effect process. The two chapters following this one—building test statistics for particular and general situations, and robust, effective model-building—lean heavily on the results of this chapter.

9.2 Context and motivation

O'Quigley (2003) developed a process based on the regression effect of a proportional or a non-proportional hazards model. In that paper it was shown how an appropriate transformation of the time scale allowed us to visualize this process in relation to standard processes such as Brownian motion and the Brownian bridge, two processes that arise as limiting results for the regression effect process

J. O'Quigley, *Survival Analysis*,
https://doi.org/10.1007/978-3-030-33439-0_9



(a) $\beta = 0$ (b) $\beta = 0.5$

Figure 9.1: The regression effect process as a function of time for simulated data under the proportional hazards model for different values of β.

under certain conditions. O'Quigley (2003) and O'Quigley (2008) indicate how we can move back to a regression effect process on the original scale if needed. While conceptually the multivariate situation is not really a greater challenge than the univariate case, the number of possibilities increases greatly. For instance, two binary covariates give rise to several processes; two marginal processes, two conditional processes, and a process based on the combined effect as expressed through the prognostic index. In addition we can construct processes for stratified models. All of these processes have known limiting distributions when all model assumptions are met. However, under departures from working assumptions, we are faced with the problem of deciding which processes provide the most insight into the collection of regression effects, taken as a whole or looked at individually (Figure 9.1).

In the presentation of this chapter we follow the outline given by O'Quigley (2003) and O'Quigley (2008). The key results that enable us to appeal to the form of Brownian motion and the Brownian bridge are proven in O'Quigley (2003), O'Quigley (2008), Chauvel and O'Quigley (2014, 2017), and Chauvel (2014). The processes of interest to us are based on cumulative quantities having the flavor of weighted residuals, i.e., the discrepancy between observations and some mean effect determined by the model. The model in turn can be seen as a mechanism that provides a prediction and this prediction can then be indexed by one or more unknown parameters. Different large sample results depend on what values we choose to use for these parameters. They may be values fixed under an assumption given by a null hypothesis or they may be some or all of the values provided by estimates from our data. In this latter case the process will tell us something about goodness of fit (O'Quigley, 2003). In the former case the process will tell us about different regression functions that can be considered to be compatible with the observations.

9.3 Elements of the regression effect process

In the most basic case of a single binary covariate, the process developed below in Section 9.4 differs from that of Wei (1984) in three very simple ways: the sequential standardization of the cumulative process, the use of a transformed time scale, and the direct interpolation that leads to the needed continuity. These three features allow for a great deal of development, both on a theoretical and practical level.

Taking the regression parameter to be β, then at time t, the process of Wei is given by

$$U(\beta,t) = \sum_{i=1}^{n} \int_0^t \left\{ Z_i(s) - \mathcal{E}_\beta(Z \mid s) \right\} dN_i(s), \quad 0 \le t \le \mathcal{T}. \tag{9.1}$$

The process corresponds to a sum which can be broken down into a series of sequential increments, each increment being the discrepancy between the observation and the expected value of this observation, conditional upon time s and the history up to that point. It is assumed that the model generates the observations. Following the last usable observation (by usable we mean that the conditional variance of the covariable in the risk set is strictly greater than zero), the resulting quantity is the same as the score obtained by the first derivative of the log likelihood before time transformation. For this reason the process is also referred to as the score process. In view of the immediate generalization of the score process and the use we will make of this process we prefer to refer to it as the regression effect process.

The point of view via empirical processes has been adopted by other authors. Arjas (1988) developed a process that is similar to that considered by Wei. Barlow and Prentice (1988) developed a class of residuals by conditioning in such a way—not unlike our own approach—that the residuals can be treated as martingales. The term martingale residuals is used. In essence any weight that we can calculate at time t, using only information available immediately prior to that time, can be used without compromising the martingale property. Such quantities can be calculated for each individual and subsequently summed. The result is a weighted process that contains the usual score process as a particular case (Lin et al., 1993; Therneau and Grambsch, 2000). The constructions of all of these processes take place within the context of the proportional hazards model. Extending the ideas to non-proportional hazards does not appear to be very easy and we do not investigate it outside of our own construction based on sequential standardization and a transformation of the time scale. Wei (1984) was interested in model fit and showed how a global, rather than a sequential standardization, could still result in a large sample result based on a Brownian bridge.

Sequential standardization and time transformation lead to great simplification, to immediate extensions to the multivariate case, and to covariate dependent censoring. The simplicity of the resulting graphics is also compelling and, finally, the simple theorems enable us to use these results in practice with confidence. Although more challenging in our view, other authors have had success in extending the results of Wei (1984). Haara (1987) was able to find analogous results in the case of non-binary covariables while Therneau et al. (1990) considered a global standardization leading to a process that would look like a Brownian bridge under the condition that the covariables were uncorrelated. This would not typically be a realistic assumption. Given the very limited number of situations, under global standardization, in which we can appeal to large sample results looking like Brownian motion or the Brownian bridge, Lin et al. (1993) adopted a different approach based on martingale simulation. This is more general and while the resulting process has no form that can be anticipated we can appeal to computer power to generate a large number of simulated processes under the appropriate hypothesis. Tracing out all of these processes, over the points of evaluation, provides us with envelopes that are indexed by their percent coverage properties. If the observed process lies well within the envelope, we may take that as evidence that any violation of the model assumptions is not too great. "Well within" is not easily defined so that, again, it seems preferable to have procedures based on more analytic results. Our own preference is for sequential rather than global standardization and time transformation since, as we see more fully below, this avoids having to deal with a lot of messy looking figures. More importantly, it provides an immediate impression as to directions in which to look if we have doubts about the underlying assumption of proportionality of hazards. Lin et al. (1996) extended this envelope approach for situations of covariate dependent censoring.

In order to give ourselves a foretaste as to where this is all pointing, consider a simple example taken from the breast cancer study at the Curie Institute in Paris. One covariable of interest was age at diagnosis categorized into two groups, the hypothesis being that the younger group may harbor more aggressive tumors. The results are illustrated in Figure 9.2. The leftmost figure shows the Kaplan-Meier curves for the two groups with some suggestion that there may be a difference. The middle figure gives the regression effect process in which the arrival point of the process is close to the value -2.0, indicating a result that would be considered to be borderline significant at 5%. Of equal importance is the form of the regression effect process which, as we will show in this chapter, will look like Brownian motion with linear drift under a non-null effect of a proportional hazards nature. In Appendix B we see how this transforms to a Brownian bridge, the 95% limits of the maximum which are shown for reference in the rightmost figure. The bridge process lies well within those limits and shows no obvious sign of departure from a Brownian bridge assumption allowing us to conclude that there is no compelling reason to not accept a working assumption of proportional

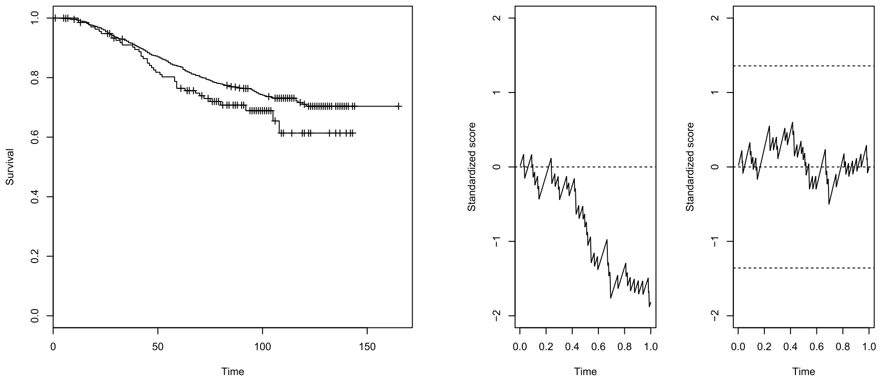


Figure 9.2: Kaplan-Meier curves for the two age groups from the Curie Institute breast cancer study. The regression effect process $U^*(t)$ approximates that of a linear drift while the transformation $U^*(t) - tU^*(1)$ of the rightmost figure appears well approximated by a Brownian bridge. This indicates a satisfactory fit.

hazards. All of this can be quantified and framed within more formal inferential structures. The key point here is that before any formal inference takes place, the eyeball factor—a quick glance at the Kaplan-Meier curves, the regression effect process, and its transform—tells us a lot.

The empirical process we construct in the following sections allows us to perform hypothesis tests on the value of the parameter and study the adequacy of the non-proportional hazards model without having to estimate the regression coefficient. In addition, increments of the process are standardized at each time point, rather than adopting an overall standardization for all increments as in Wei (1984). This makes it possible to take into account multiple and correlated covariates because, unlike the non-standardized score process (9.1), the correlation between covariates is not assumed to be constant over time. We plot the process according to ranked times of failure, in the manner of Arjas (1988), to obtain a simple and explicit asymptotic distribution. Before constructing the process, we need to change the time scale. This transformation is crucial for obtaining asymptotic results for the process without having to use the theorem of Rebolledo (1980), nor the inequality of Lenglart (1977), which are classical tools in survival analysis. In the following section, we present the time transformation used. In Section 9.4, we define the univariate regression effect process and study its asymptotic behavior. In Section 9.5, we look at the multivariate case. Section 9.8 gathers together the proofs of the theorems and lemmas that are presented.

Transforming the time scale

In the proportional hazards model, a strictly monotonically increasing transformation in time does not affect inference on the regression coefficient β. Indeed, the actual times of death and censored values are not used for inference; only their order of occurrence counts. Among all possible monotonic increasing transformations we choose one of particular value. The order of observed times is preserved and, by applying the inverse transformation, we can go back to the initial time scale. Depending on how we carry out inference, there can be a minor, although somewhat sticky, technical difficulty, in that the number of distinct, and usable, failure times, depends on the data and is not known in advance. Typically we treat this number as though it were known and, indeed, our advice is to do just that in practice. This amounts to conditioning on the number of distinct and usable failure times. We use the word "usable" to indicate that information on regression effects is contained in the observations at these time points. In the case of 2 groups for example, once one group has been extinguished, then, all failure times greater than the greatest for this group carry no information on regression effects.

Definition 9.1. *With this in mind we define the effective sample size as*

$$k_n = \sum_{i=1}^{n} \delta_i \times \Delta(X_i) \quad \textit{where} \quad \Delta(t) = \mathbf{1}_{\mathcal{V}_{\beta(t)}(Z|t)>0}$$

If the variance is zero at a time of death, this means that the set of individuals at risk is composed of a homogeneous population sharing the same value of the covariate. Cleary the outcome provides no information on differential rates as quantified by the regression parameter β. This translates mathematically as zero contribution of these times of death to inference, since the value of the covariate of the individual who dies is equal to its expectation.

We want to avoid such variances from a technical point of view, because later we will want to normalize by the square root of the variance. For example, in the case of a comparison of two treatment groups, a null conditional variance corresponds to the situation in which one group is now empty but individuals from the other group are still present in the study. Intuitively, we understand that these times of death do not contribute to estimation of the regression coefficient β since no information is available to compare the two groups. Note that the nullity of a conditional variance at a given time implies nullity at later time points. Thus, according to the definition of k_n, the conditional variances $\mathcal{V}_{\beta(t)}(Z|t)$ calculated at the first k_n times of death t are strictly greater than zero. For a continuous covariate when the last time point observed is a death, we have the equality (algebraic if conditioning, almost sure otherwise) that $k_n = \sum_{i=1}^{n} \delta_i - 1$. In effect, the conditional variance $\mathcal{V}_{\beta(t)}(Z|t)$ is zero at the last observed time point t, and almost surely is the only time at which the set of at-risk individuals shares the same value of the covariate. If, for a continuous covariate, the last observed time point is censored, then we have that $k_n = \sum_{i=1}^{n} \delta_i$.

Definition 9.2. *Set $\phi_n(0) = 0$ and consider the following transformation ϕ_n of the observed times $X_i, i = 1,...,n$:*

$$\phi_n(X_i) = \frac{1}{k_n}\left[\bar{N}(X_i) + (1-\delta_i)\frac{\#\left\{j : j = 1,\ldots,n, X_j < X_i,\ \bar{N}(X_j) = \bar{N}(X_i)\right\}}{\#\left\{j : j = 1,\ldots,n,\ \bar{N}(X_j) = \bar{N}(X_i)\right\}}\right]$$

We define the inverse ϕ_n^{-1} of ϕ_n on $[0,1]$ by

$$\phi_n^{-1}(t) = \inf\{X_i,\ \phi_n(X_i) \ge t,\ i = 1,\ldots,n\}, \quad 0 \le t \le 1.$$

Recall that the counting process $\{\bar{N}(t)\}_{t\in[0,\mathcal{T}]}$ has a jump of size 1 at each time of death. Thus, on the new time scale representing the image of the observed times $\{X_1,\ldots,X_n\}$ under ϕ_n, the values in the set $\{1/k_n, 2/k_n,\ldots,1\}$ correspond to times of death, and the ith time of death t_i is such that $t_i = i/k_n, \quad i = 1,\ldots,k_n$ where $t_0 = 0$. The set $\{1/k_n, 2/k_n,\ldots,1\}$ is included in but is not necessarily equal to the transformed set of times of death.

Inference based on transformed and original time scale

Inference for the regression coefficient is unaffected by monotonic increasing transformations on the time scale and, in particular, the transformation of Definition 9.2. The nature of effects will be very visible on the transformed time scale and less so on the original scale. Nonetheless we can use the transform, and its inverse, to move back and forward between scales. The transformation, $\phi_n(X_i)$, is not independent of the censoring and so some caution is needed in making statements that we wish to hold regardless of the censoring. A regression effect process with linear drift implies and is implied by proportional hazards. Constancy of effect on one scale will then imply constancy of effect on the other. We can say more. Suppose we have a changepoint model. The ratio of the early regression coefficient to the later regression coefficient will be the same on either scale.

Working with the first k_n of the transformed failures, we will therefore restrict ourselves to values less than or equal to 1. As the importance of the transformation applied to the observed times is to preserve order, various transformations—treating censoring times differently—could be used. We choose to uniformly distribute censoring times between adjacent times of death. The time τ_0 on this new scale corresponds to the $(100 \times \tau_0)$th percentile for deaths in the sample. For example, at time $\tau_0 = 0.5$, half of the deaths have been observed. At time $\tau_0 = 0.2$, 20% of the deaths have occurred. The inverse of ϕ_n can be obtained easily and with it we can interpret the results on the initial time scale.

Conditional means and variances

Our next task is that of sequential standardization. We now work with the standardized time scale in $[0,1]$. With this time scale, for $t \in [0,1]$, we define the hazard index $Y_i^*(t)$ and individual counting process $N_i^*(t)$ for individuals $i = 1,\ldots,n$ by

$$Y_i^*(t) = \mathbf{1}_{\phi_n(X_i) \geq t}, \qquad N_i^*(t) = \mathbf{1}_{\phi_n(X_i) \leq t, \delta_i = 1}.$$

Each time of death t_i is a $(\mathcal{F}_t^*)_{t\in[0,1]}$-stopping time where, for $t \in [0,1]$, the σ-algebra $\mathcal{F}_t^*$ is defined by

$$\mathcal{F}_t^* = \sigma\left\{N_i^*(u), Y_i^*(u^+), Z_i\,; i = 1,\ldots,n\,; u = 0,1,\ldots,\lfloor tk_n \rfloor\right\},$$

where $\lfloor\cdot\rfloor$ is the floor function and $Y_i^*(t^+) = \lim_{s\to t^+} Y_i^*(s)$. Notice that if $0 \leq s < t \leq 1$, $\mathcal{F}_s^* \subset \mathcal{F}_t^*$.

Remark. Denote $a \wedge b = \min(a,b)$, for $a, b \in \mathbb{R}$. If the covariates are time-dependent, for $t \in [0,1]$, the σ-algebra $\mathcal{F}_t^*$ is defined by

$$\mathcal{F}_t^* = \sigma\left\{N_i^*(u), Y_i^*(u^+), Z_i\left(\phi_n^{-1}\left(\frac{u}{k_n}\right)^+ \wedge X_i\right); i = 1,\ldots,n\,; u = 0,1,\ldots,\lfloor tk_n\rfloor\right\}$$

since, as we recall, the covariate $Z_i(\cdot)$ is not observed after time X_i, $i = 1,\ldots,n$. To simplify notation in the following, we will write $Z_i\left(\phi_n^{-1}(t)\right)$ in the place of $Z_i\left(\phi_n^{-1}(t)^+ \wedge X_i\right)$, , $t \in [0,1]$. We define the counting process associated with transformed times, with jumps of size 1 at times of death in the new scale, by

$$\bar{N}^*(t) = \sum_{i=1}^n \mathbf{1}_{\phi_n(X_i) \leq t, \delta_i = 1}, \quad 0 \leq t \leq 1.$$

This counting process satisfies the following result:

Proposition 9.1. (Chauvel 2014). $\forall t \in [0,1]$, $\bar{N}^*(t) = \lfloor k_n t \rfloor$.

In other words, $t_0 = 0$ and $\bar{N}^*(t)$ has jumps at times $t_j = j/k_n$ for $j = 1,\ldots,k_n$. This leads to the following lemma, which will be of use in proving Theorem 9.2:

Lemma 9.1. (Chauvel and O'Quigley 2014). *Let $t_0 = 0$, $n \in \mathbb{N}$ and let $\{A_n(t), t \in \{t_1,\ldots,t_{k_n}\}\}$ be a process with values in $\mathbb{R}$. Suppose that*

$$\sup_{i=1,\ldots,k_n} |A_n(t_i) - a(t_i)| \quad \xrightarrow[n\to\infty]{P} \quad 0,$$

where a is a function defined and bounded on $[0,1]$. Then,

$$\sup_{s\in[0,1]}\left|\frac{1}{k_n}\int_0^s A_n(t)d\bar{N}^*(t) - \int_0^s a(t)dt\right| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

The proofs are given at the end of the chapter. To ease notation, let us define a process $\mathcal{Z}$ which is 0 everywhere except at times of death. At each such time, the process takes the value of the covariate of the individual who fails at that time.

Definition 9.3. *We denote $\mathcal{Z} = \{\mathcal{Z}(t),\ t\in[0,1]\}$ a process such that*

$$\mathcal{Z}(t) = \sum_{i=1}^n Z_i\left(X_i\right)\mathbf{1}_{\phi_n(X_i)=t,\,\delta_i=1}, \quad t\in[0,1]. \tag{9.2}$$

The family of probabilities $\{\pi_i(\beta(t),t), i = 1,\dots,n\}$, with $t\in[0,\mathcal{T}]$ can be extended to a non-proportional hazards model on the transformed scale as follows.

Definition 9.4. *For $i=1,\dots,n$ and $t\in[0,1]$, the probability that individual i dies at time t under the non-proportional hazards model with parameter $\beta(t)$, conditional on the set of individuals at risk at time of death t, is defined by*

$$\pi_i(\beta(t),t) = \frac{Y_i^*(t)\exp\left(\beta(t)^T Z_i\left(\phi_n^{-1}(t)\right)\right)\mathbf{1}_{k_n t\in\mathbb{N}}}{\sum_{j=1}^n Y_j^*(t)\exp\left(\beta(t)^T Z_j\left(\phi_n^{-1}(t)\right)\right)}.$$

Remark. The only values at which these quantities will be calculated are the times of death $t_1,\dots,t_{k_n}$. We nevertheless define $\mathcal{Z}(t)$ and $\{\pi_i(\beta(t),t), i = 1,\dots,n\}$ for all $t\in[0,1]$ in order to be able to write their respective sums as integrals with respect to the counting process $\bar{N}^*$. The values 0 taken by them at times other than that of death are chosen arbitrarily as long as they remain bounded by adjacent failures. Note also that at $t_0 = 0$, we have $\mathcal{Z}(0) = 0$. Expectations and variances with respect to the family of probabilities $\{\pi_i(\beta(t),t), i=1,\dots,n\}$ can now be defined.

Definition 9.5. *Let $t\in[0,1]$. The expectation and variance of Z with respect to the family of probabilities $\{\pi_i(\beta(t),t), i=1,\dots,n\}$ are respectively a vector in $\mathbb{R}^p$ and a matrix in $\mathcal{M}_{p\times p}(\mathbb{R})$ such that*

$$\mathcal{E}_{\beta(t)}(Z\mid t) = \sum_{i=1}^n Z_i\left(\phi_n^{-1}(t)\right)\pi_i(\beta(t),t), \tag{9.3}$$

and

$$\mathcal{V}_{\beta(t)}(Z\mid t) = \sum_{i=1}^n Z_i\left(\phi_n^{-1}(t)\right)^{\otimes 2}\pi_i(\beta(t),t) - \mathcal{E}_{\beta(t)}(Z\mid t)^{\otimes 2}. \tag{9.4}$$

It can be shown that the Jacobian matrix of $\mathcal{E}_{\beta(t)}(Z \mid t)$ *is* $\dfrac{\partial}{\partial \beta}\mathcal{E}_{\beta(t)}(Z \mid t) = \mathcal{V}_{\beta(t)}(Z \mid t)$*, for* $t \in [0,1]$*.*

Proposition 9.2. *Let* $t \in [0,1]$*. Under the non-proportional hazards model with parameter* $\beta(t)$*, the expectation* $E_{\beta(t)}(\mathcal{Z}(t) \mid \mathcal{F}^*_{t-})$ *and variance* $V_{\beta(t)}(\mathcal{Z}(t) \mid \mathcal{F}^*_{t-})$ *of the random variable* $\mathcal{Z}(t)$ *given the* σ*-algebra* $\mathcal{F}^*_{t-}$ *are*

$$E_{\beta(t)}(\mathcal{Z}(t) \mid \mathcal{F}^*_{t-}) = \mathcal{E}_{\beta(t)}(Z \mid t), \quad V_{\beta(t)}(\mathcal{Z}(t) \mid \mathcal{F}^*_{t-}) = \mathcal{V}_{\beta(t)}(Z \mid t).$$

In conclusion, for $j = 1,\ldots,k_n$ and time of death t_j, by conditioning on the set of at-risk individuals, the expectation of $\mathcal{Z}(t_j)$ is $\mathcal{E}_{\beta(t_j)}(Z \mid t_j)$ and its variance-covariance matrix is $\mathcal{V}_{\beta(t_j)}(Z \mid t_j)$. Knowledge of these moments allows us to define the regression effect process, which we will study separately for the univariate $(p = 1)$ and multivariate $(p > 1)$ cases. It is worth adding that the univariate process has an interest well beyond the case of simple models involving only a single covariate. For multivariate models there are several univariate processes that will reveal much of what is taking place. These include the process for the prognostic index, processes for individual covariates after having adjusted via the model for the effects of some or all of the other covariates, processes for stratified models, and indeed any process that can be derived from the model.

9.4 Univariate regression effect process

In this section, we will focus more closely on the univariate framework $(p = 1)$, i.e., when there is only one covariate $Z(t)$ in the model. In some sense higher dimensions represent an immediate generalization of this but, for higher dimensions, in view of the complexity theorem (Theorem 4.1) there are many more things to consider, both conceptually and operationally. In order to gain the firmest grip possible on this we look closely at the univariate case which contains all of the key ideas. It is always a good principle in statistics to condition on as many things as we can. We can go too far and condition on aspects of the data that are not uninformative with regard to our unknown regression parameter. This would certainly lead to error so care is always called for. In some ways conditioning is an art, knowing just how far to go. One thing that can guide us is to what extent, after such conditioning, we can recover procedures that have been established under quite different principles. Here, we will see for instance that a lot of conditioning will result in the well-known log-rank test which, at the very least, gives us some confidence.

Sequential conditioning through time, i.e., conditioning on the times of death and the sets of individuals at risk, as first proposed by Cox (1972), leads to a very natural framework for inference. An individual i who dies at t_j is viewed as having been selected randomly (under the model) from the set

$\{i : Y_i^*(t_j) = 1;\ i = 1,\ldots,n\}$ of individuals at risk of death at time t_j with probability $\pi_i(\beta(t_j),t_j)$. Under this conditioning, at the time of death t_j, the information necessary to calculate the expectation and variance of $\mathcal{Z}(t_j)$ with respect to $\{\pi_i(\beta(t_j),t_j);\ i=1,\ldots,n\}$ is available for $j=1,\ldots,k_n$. The conditional variances $\mathcal{V}_{\beta(t)}(Z \mid t)$ calculated for the first k_n times of death t are strictly greater than zero. Thus,

$$\forall t \in \{t_1,\ldots,t_{k_n}\},\ \mathcal{V}_{\beta(t)}(Z \mid t) > 0 \quad \text{a.s.}$$

The variables $\mathcal{Z}(t_j)$ may be standardized. By sequentially summing all of these standardized variables, we obtain a standardized score which is the basis of our regression effect process. A further conditioning argument whereby the number of discrete failure times at which the variance $\mathcal{V}_{\beta(t)}(Z \mid t)$ is strictly greater than zero leads, once more, to great simplification. For completeness the unconditioned case is worked through at the end of the chapter. As argued above, the most useful approach is a conditional one in which we take k_n to be equal to its observed value.

Definition 9.6. *Let $j = 0,1,\ldots,k_n$. The regression effect process evaluated for the parameter $\beta(t_j)$ at time t_j is defined by $U_n^*(\cdot,\cdot,0) = 0$ and*

$$U_n^*\left(\alpha(t_j),\beta(t_j),t_j\right) = \frac{1}{\sqrt{k_n}}\int_0^{t_j} \mathcal{V}_{\alpha(s)}(Z \mid s)^{-1/2}\left\{\mathcal{Z}(s) - \mathcal{E}_{\beta(s)}(Z \mid s)\right\} d\bar{N}^*(s).$$

where, by using two parameter functions, $\alpha(t)$ and $\beta(t)$, we allow ourselves increased flexibility. The $\alpha(t)$ concerns the variance and allowing this to not be tied to $\beta(t)$ may be of help in some cases where we would like estimates of the variance to be valid both under the null and the alternative. This would be similar to what takes place in an analysis of variance where the residual variance is estimated in such a way as to remain consistent both under the null and the alternative. Our study on this, in this particular context, is very limited, and this could be something of an open problem. For the remainder of this text we will suppose that $\alpha(t) = \beta(t)$ and we will consequently write $U_n^*(\beta(t_j),t_j)$ as only having two arguments.

Remark. For computational purposes, and for $j = 1,\ldots,k_n$, the variable $U_n^*(\beta(t_j),t_j)$ can also be written as the sum:

$$U_n^*(\beta(t_j),t_j) = \frac{1}{\sqrt{k_n}}\sum_{i=1}^{j} \mathcal{V}_{\beta(t_i)}(Z \mid t_i)^{-1/2}\left\{\mathcal{Z}(t_i) - \mathcal{E}_{\beta(t_i)}(Z \mid t_i)\right\}, \qquad (9.5)$$

noting that all of the ingredients in the above expression are obtained routinely from all of the currently available software packages.

Remark. Let w be a deterministic or random $(\mathcal{F}_t^*)_{t\in[0,1]}$-predictable function from $[0,1]$ to $\mathbb{R}$. According to Proposition 9.2, under the non-proportional hazards model with parameter $\beta(t)$, conditional on the set of individuals at risk at time of death t_j, the expectation of $w(t_j)\mathcal{Z}(t_j)$ is $w(t_j)\mathcal{E}_{\beta(t_j)}(Z \mid t_j)$ and its variance $w(t_j)^2\mathcal{V}_{\beta(t_j)}(Z \mid t_j)$ with respect to the family of probabilities $\{\pi_i(\beta(t_j),t_j);\ i=1,\dots,n\}$. By summing the standardized weighted variables, we obtain the same definition of the process as in Definition 9.6. We will come back to this remark in Section 11.4 for the weighted log-rank test.

We can now define the process U_n^* on $[0,1]$ by linearly interpolating the k_n+1 random variables $\{U_n^*(\beta(t_j),t_j),\ j=0,1,\dots,k_n\}$.

Definition 9.7. *The standardized score process evaluated at $\beta(t)$ is defined by $\{U_n^*(\beta(t),t),\ t\in[0,1]\}$, where for $j=0,\dots,k_n$ and $t\in[t_j,t_{j+1}[$,*

$$U_n^*(\beta(t),t) = U_n^*(\beta(t_j),t_j) + (tk_n - j)\left\{U_n^*(\beta(t_{j+1}),t_{j+1}) - U_n^*(\beta(t_j),t_j)\right\}.$$

By definition, the process $U_n^*(\beta(\cdot),\cdot)$ is continuous on the interval $[0,1]$. The process depends on two parameters: the time t and regression coefficient $\beta(t)$. For the sake of clarity, we recall that the temporal function $\beta : t \to \beta(t)$ is denoted $\beta(t)$, which will make it easier to distinguish the proportional hazards model with parameter β from the non-proportional hazards model with parameter $\beta(t)$. Under the non-proportional hazards model with $\beta(t)$, the increments of the process U_n^* are centered with variance 1. Moreover, the increments are uncorrelated, as shown in the following proposition.

Proposition 9.3. (Cox 1975). *Let $D_n^*(\beta(t_j),t_j) = U_n^*(\beta(t_j),t_j) - U_n^*(\beta(t_{j-1}), t_{j-1})$. Under the non-proportional hazards model with parameter $\beta(t)$, the random variables $D_n^*(\beta(t_j),t_j)$ for $j=1,2,\dots,k_n$ are uncorrelated.*

The uncorrelated property of the increments is also used in the calculation of log-rank statistic. All of these properties together make it possible to obtain our essential convergence results for the process $U_n^*(\beta(t),t)$.

WORKING ASSUMPTIONS FOR LARGE SAMPLE THEORY

The space $(D[0,1],\mathbb{R})$ comes equipped with uniform convergence. For $0\le t\le 1$ and for $r=0,1,2$, we can define:

$$S^{(r)}\{\beta(t),t\} = \frac{1}{n}\sum_{i=1}^{n} Y_i^*(t) Z_i\left(\phi_n^{-1}(t)\right)^r \exp\left(\beta(t) Z_i\left(\phi_n^{-1}(t)\right)\right).$$

Note that for $t\in\{t_1,t_2,\dots,t_{k_n}\}$,

$$\mathcal{E}_{\beta(t)}(Z \mid t) = \frac{\partial}{\partial b}\log\left\{S^{(0)}(b,t)\right\}\bigg|_{b=\beta(t)} = \frac{S^{(1)}(\beta(t),t)}{S^{(0)}(\beta(t),t)},$$

and

$$\mathcal{V}_{\beta(t)}(Z \mid t) = \frac{\partial^2}{\partial b^2} \log \left\{ S^{(0)}(b,t) \right\} \Big|_{b=\beta(t)} = \frac{S^{(2)}(\beta(t),t)}{S^{(0)}(\beta(t),t)} - \left(\frac{S^{(1)}(\beta(t),t)}{S^{(0)}(\beta(t),t)} \right)^2 .$$

Suppose now the following conditions:

A1 (Asymptotic stability). There exists some $\delta_1 > 0$, a neighborhood $\mathbb{B} = \left\{ \gamma, \sup_{t \in [0,1]} |\gamma(t) - \beta(t)| < \delta_1 \right\}$ of β of radius δ_1 containing the zero function, and functions $s^{(r)}$ defined on $\mathbb{B} \times [0,1]$ for $r = 0,1,2$, such that

$$\sqrt{n} \sup_{t \in [0,1], \gamma \in \mathbb{B}} \left| S^{(r)}(\gamma(t),t) - s^{(r)}(\gamma(t),t) \right| \quad \xrightarrow[n \to \infty]{P} \quad 0. \tag{9.6}$$

A2 (Asymptotic regularity). The deterministic functions $s^{(r)}$, $r = 0,1,2$, defined in A1 are uniformly continuous for $t \in [0,1]$ and bounded in $\mathbb{B} \times [0,1]$. Furthermore, for $r = 0,1,2$, and $t \in [0,1]$, $s^{(r)}(\cdot,t)$ is a continuous function in $\mathbb{B}$. The function $s^{(0)}$ is bounded below by a strictly positive constant.

By analogy with the empirical quantities, for $t \in [0,1]$ and $\gamma \in \mathbb{B}$, denote

$$e(\gamma(t),t) = \frac{s^{(1)}(\gamma(t),t)}{s^{(0)}(\gamma(t),t)}, \quad v(\gamma(t),t) = \frac{s^{(2)}(\gamma(t),t)}{s^{(0)}(\gamma(t),t)} - e(\gamma(t),t)^2. \tag{9.7}$$

A3 (Homoscedasticity). For any $t \in [0,1]$ and $\gamma \in \mathbb{B}$, we have $\dfrac{\partial}{\partial t} v(\gamma(t),t) = 0.$

A4 (Uniformly bounded covariates). There exists $L \in \mathbb{R}^{*+}$ such that

$$\sup_{i=1,\dots,n} \sup_{u \in [0,\mathcal{T}]} |Z_i(u)| \le L.$$

The first two conditions are standard in survival analysis. They were introduced by Andersen and Gill (1982) in order to use theory from counting processes and martingales such as the inequality of Lenglart (1977) and the theorem of Rebolledo (1980). The variance $\mathcal{V}_{\beta(t)}(Z \mid t)$ is, by definition, an estimator of the variance of Z given $T = t$ under the non-proportional hazards model with parameter $\beta(t)$. Thus, the homoscedasticity condition A3 means that the asymptotic variance is independent of time for parameters close to the true regression coefficient $\beta(t)$.

This condition is often implicitly encountered in the use of the proportional hazards model. This is the case, for example, in the estimation of the variance of the parameter β, or in the expression for the log-rank statistic, where the contribution to the overall variance is the same at each time of death. Indeed, the overall variance is an unweighted sum of the conditional variances. The temporal stability of the variance has been pointed out by several authors, notably Grambsch and Therneau (1994), Xu (1996), and Xu and O'Quigley (2000). Next, we need two lemmas. Proofs are given at the end of the chapter.

Lemma 9.2. *Under conditions A1 and A3, for all $t \in [0,1]$ and $\gamma \in \mathbb{B}$, $v(\gamma(t),t) > 0$.*

Lemma 9.3. *Under hypotheses, A1, A2, and A3, and if $k_n = k_n(\beta_0)$, for all $\gamma_1 \in \mathbb{B}$, there exist constants $C(\gamma_1)$ et $C(\beta_0) \in \mathbb{R}^{*+}$ such that*

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}} \left|\mathcal{V}_{\gamma_1(t)}(Z \mid t) - C(\gamma_1)\right| \quad \xrightarrow[n\to\infty]{P} \quad 0, \tag{9.8}$$

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}} \left|\frac{\mathcal{V}_{\gamma_1(t)}(Z \mid t)}{\mathcal{V}_{\beta_0}(Z \mid t)} - \frac{C(\gamma_1)}{C(\beta_0)}\right| \quad \xrightarrow[n\to\infty]{P} \quad 0, \tag{9.9}$$

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}} \left|\frac{\mathcal{V}_{\gamma_1(t)}(Z \mid t)}{\mathcal{V}_{\beta_0}(Z \mid t)^{1/2}} - \frac{C(\gamma_1)}{C(\beta_0)^{1/2}}\right| \quad \xrightarrow[n\to\infty]{P} \quad 0. \tag{9.10}$$

Theorem 9.1. *Under the non-proportional hazards model with parameter $\beta(t)$,*

$$U_n^*(\beta(\cdot),\cdot) \quad \xrightarrow[n\to+\infty]{\mathcal{L}} \quad \mathcal{W}(t),$$

where $\mathcal{W}(t)$ is a standard Brownian motion.

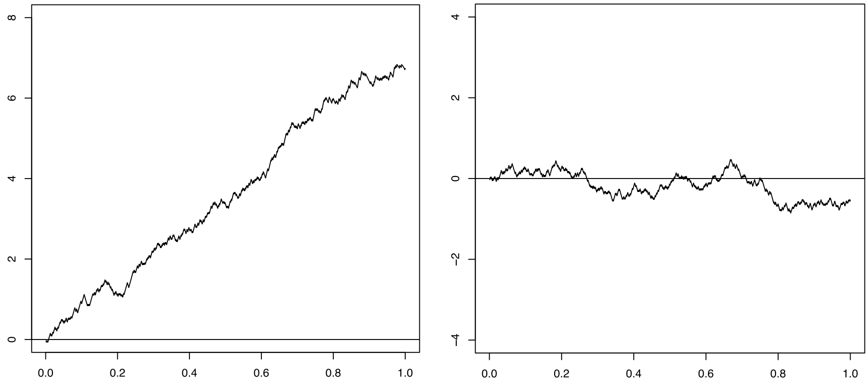
The proof, given at the end of the chapter, is built on the use of the functional central limit theorem for martingale differences introduced by Helland (1982). This theorem can be seen as an extension of the functional central limit theorem of Donsker (1951) to standardized though not necessarily independent nor identically distributed variables. Figure 9.3(a) shows the result of simulating a standardized score process $U_n^*(0,t)$ as a function of time t, when the true parameter is constant. The immediate visual impression would lead us to see Brownian motion—standard or with linear drift—as a good approximation. We will study how to calculate this approximation on a given sample in Chapters 10 and 11.

Theorem 9.2. (Chauvel and O'Quigley 2014)**.** *Let β_0 be a constant function on $\mathbb{B}$. Under the non-proportional hazards model with parameter $\beta(t)$, consider*

$$A_n(t) = \frac{1}{k_n}\int_0^t \{\beta(s) - \beta_0\} \frac{\mathcal{V}_{\tilde{\beta}(s)}(Z \mid s)}{\mathcal{V}_{\beta_0}(Z \mid s)^{1/2}} d\bar{N}^*(s), \quad 0 \le t \le 1, \tag{9.11}$$

where $\tilde{\beta}$ is in the ball with center β and radius $\sup_{t\in[0,1]} |\beta(t) - \beta_0|$. Then, there exist two constants $C_1(\beta,\beta_0)$ and $C_2 \in \mathbb{R}^{*+}$ such that $C_1(\beta,\beta)$=1 and

$$U_n^*(\beta_0,\cdot) - \sqrt{k_n}A_n \quad \xrightarrow[n\to\infty]{\mathcal{L}} \quad C_1(\beta,\beta_0)\mathcal{W}(t). \tag{9.12}$$



(a) $\beta = 1.0$ (b) $\beta = 0.0$

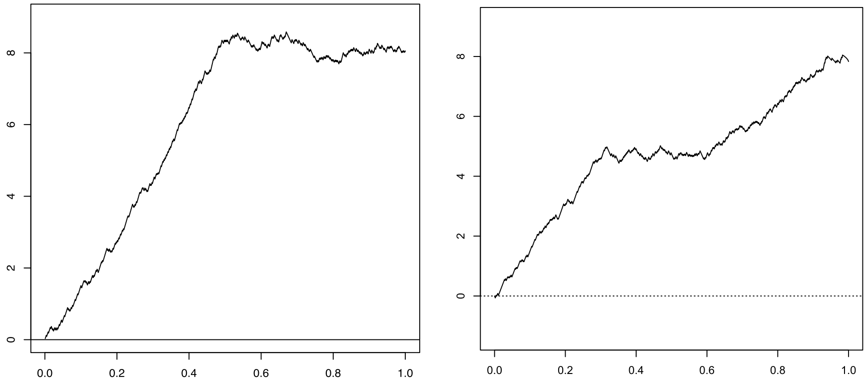
Figure 9.3: The process $U_n^*(0,\cdot)$ as a function of time for simulated data under the proportional hazards model. For left-hand figure, $\beta = 1.0$. For right-hand figure, the absence of effects manifests itself as approximate standard Brownian motion.

Furthermore,

$$\sup_{0\le t\le 1}\left|A_n(t) - C_2\int_0^t \{\beta(s)-\beta_0\}\,ds\right| \quad \xrightarrow[n\to\infty]{P} \quad 0. \tag{9.13}$$

Remark. Theorem 9.1 is a special case of Theorem 9.2 when $\beta(t) = \beta_0$. In effect, in Theorem 9.2, when $\beta(t) = \beta_0$, the constant $C_1(\beta,\beta)$ is equal to 1, the drift is zero, and the limit process of $U_n^*(\beta_0,\cdot)$ is a standard Brownian motion. The proof of the theorem can be found in Section 9.8. The drift of the standardized score process at time t tends to infinity when $k_n \to \infty$, according to equation (9.12). In practice, we work with a large enough fixed k_n and $\beta_0 = 0$. The theorem indicates that the drift of the process $U_n^*(0,\cdot)$ is proportional to $\beta(s)$. We will distinguish between two cases in which the regression coefficient $\beta(t)$ is non-zero. In the first, consider the proportional hazards model, that is, $\beta(t)$ is constant over time. From Theorem 9.2, the process $U_n^*(0,\cdot)$ can be approximated by a Brownian motion with a linear drift. This is illustrated in Figure 9.3(a) with a process that has been simulated under the proportional hazards model with non-zero β. The drift can be approximated by a straight line, indicating that the regression coefficient is constant in time. The coefficient β, which has the same sign as the slope of the line according to Theorem 9.2, is positive.

The second case corresponds to the situation where there is a non-constant regression coefficient over time. Several scenarios are possible. For example, the effect could be constant for an initial period and then quickly decline beyond



(a) $\beta(t) = \mathbf{1}_{t\leq 0.5}$

(b) $\beta(t) = \mathbf{1}_{t\leq 1/3} + 0.5\,\mathbf{1}_{t\geq 2/3}$

Figure 9.4: The process $U_n^*(0,\cdot)$ as a function of time for simulated data under the non-proportional hazards model with different values of $\beta(t)$. Temporal effects are clearly reflected in the process on the transformed time scale.

some point τ, piecewise constant over time, increase or decrease continuously, and so on. In all these situations, the cumulative effect $\int \beta(s)ds$ will be seen in the drift of the process (Theorem 9.2) and will, of itself, describe to us the nature of the true effects. For example, Figure 9.4(a) shows a simulated process with a piecewise-constant effect $\beta(t) = \mathbf{1}_{t\leq 0.5}$. Before $t = 0.5$, we see a positive linear trend corresponding to $\beta = 1$, and for $t > 0.5$, the coefficient becomes zero and the drift of the process $U_n^*(0,\cdot)$ is parallel to the time axis. Figure 9.4(b) shows the standardized score process for simulated data under failure time model with regression function, $\beta(t) = \mathbf{1}_{t\leq 1/3} + 0.5\,\mathbf{1}_{t\geq 2/3}$. The drift of the process can be separated into three linear sections reflecting the effect's strength, each of which can be approximated by a straight line. The slope of the first line appears to be twice that of the last, while the slope of the second line is zero.

In summary, whether the effect corresponds to proportional or non-proportional hazards, all information about the regression coefficient $\beta(t)$ can be found in the process $U_n^*(0,\cdot)$. We can then use this process to test the value of the regression coefficient and evaluate the adequacy of the proportional hazards model. A simple glance at the regression effect process is often all that is needed to let us know that there exists an effect and the nature of that effect. More precise inference can then follow. Before delving into these aspects in Chapters 10 and 11, we consider more closely the standardized score process for the multivariate case. As mentioned earlier the univariate process, even in the multivariate setting, will enable us to answer most of the relevant questions, such as the behavior of many variables combined in a prognostic index, or, for example, the impact of some variable after having controlled for the effects of other variables, whether via the model or via stratification.

9.5 Regression effect processes for several covariates

Here we extend the regression effect process of the previous section to the case of several covariates grouped in a vector $Z(t)$ of dimension $p > 1$. Suppose that we have the non-proportional hazards model with regression function $\beta(t)$ and a vector of covariates $Z(t)$ which are functions from $[0,1]$ to $\mathbb{R}^p$. Following Chauvel (2014), let $t \in [0,1]$ and

$$S^{(0)}(\beta(t),t) = n^{-1}\sum_{i=1}^{n} Y_i^*(t)\exp\left(\beta(t)^T Z_i(\phi_n^{-1}(t))\right) \in \mathbb{R},$$
$$S^{(1)}(\beta(t),t) = n^{-1}\sum_{i=1}^{n} Y_i^*(t) Z_i(\phi_n^{-1}(t))\exp\left(\beta(t)^T Z_i(\phi_n^{-1}(t))\right) \in \mathbb{R}^p,$$
$$S^{(2)}(\beta(t),t) = n^{-1}\sum_{i=1}^{n} Y_i^*(t) Z_i(\phi_n^{-1}(t))^{\otimes 2}\exp\left(\beta(t)^T Z_i(\phi_n^{-1}(t))\right) \in \mathcal{M}_{p\times p}(\mathbb{R}).$$

For $t \in \{t_1,\ldots,t_{k_n}\}$, the conditional expectation $\mathcal{E}_{\beta(t)}(Z \mid t)$ and the conditional variance-covariance matrix $\mathcal{V}_{\beta(t)}(Z \mid t)$ defined in (9.3) and (9.4) can be written as functions of $S^{(r)}(\beta(t),t)$ $(r = 0,1,2)$:

$$\mathcal{E}_{\beta(t)}(Z \mid t) = \frac{S^{(1)}(\beta(t),t)}{S^{(0)}(\beta(t),t)}, \quad \mathcal{V}_{\beta(t)}(Z \mid t) = \frac{S^{(2)}(\beta(t),t)}{S^{(0)}(\beta(t),t)} - \mathcal{E}_{\beta(t)}(Z \mid t)^{\otimes 2}.$$

which is the same as $\partial \mathcal{E}_{\beta(t)}(Z \mid t)/\partial\beta$. As the matrix $\mathcal{V}_{\beta(t)}(Z \mid t)$ is symmetric and positive definite, there exists an orthogonal matrix $P_{\beta(t)}(t)$ and a diagonal matrix $D_{\beta(t)}(t)$ such that

$$\mathcal{V}_{\beta(t)}(Z \mid t) = P_{\beta(t)}(t) D_{\beta(t)}(t) P_{\beta(t)}(t)^T.$$

This leads us to define the symmetric matrix $\mathcal{V}_{\beta(t)}(Z \mid t)^{-1/2}$ as

$$\mathcal{V}_{\beta(t)}(Z \mid t)^{-1/2} = P_{\beta(t)}(t) D_{\beta(t)}(t)^{-1/2} P_{\beta(t)}(t)^T.$$

By the definition of $\widehat{k_n} = \widehat{k_n}(\beta)$ in the multivariate case, and since $k_n \le \widehat{k_n}$ almost surely (Equation (9.23)), the conditional variance-covariance matrices $\mathcal{V}_{\beta(t)}(Z \mid t)$ calculated for the first k_n times of death t are positive definite, and

$$\forall t \in \{t_1,\ldots,t_{k_n}\},\ \left\|\mathcal{V}_{\beta(t)}(Z \mid t)^{-1}\right\| \le C_{\mathcal{V}}^{-1} \quad \text{a.s.}$$

Definition 9.8. *Let $j = 0,1,\ldots,k_n$. The multivariate standardized score calculated at time t_j for parameter $\beta(t_j)$ is defined by*

$$U_n^*(\beta(t_j),t_j) = \frac{1}{\sqrt{k_n}}\int_0^{t_j} \mathcal{V}_{\beta(s)}(Z \mid s)^{-1/2}\left\{\mathcal{Z}(s) - \mathcal{E}_{\beta(s)}(Z \mid s)\right\} d\bar{N}^*(s),$$

where

$$\int_0^{t_j} a(s)d\bar{N}^*(s) = \left(\int_0^{t_j} a_1(s)d\bar{N}^*(s), \ldots, \int_0^{t_j} a_p(s)d\bar{N}^*(s)\right),$$

for $a = (a_1, \ldots, a_p)$*, where the* a_i *are functions from* $[0,1]$ *to* $\mathbb{R}$*.*

Remark. As in the univariate case, the random variable $U_n^*(\beta(t_j), t_j)$ can be written as a sum:

$$U_n^*(\beta(t_j), t_j) = \frac{1}{\sqrt{k_n}} \sum_{i=1}^{j} \mathcal{V}_{\beta(t_i)}(Z \mid t_i)^{-1/2} \left\{\mathcal{Z}(t_i) - \mathcal{E}_{\beta(t_i)}(Z \mid t_i)\right\}.$$

The standardized score is a random function defined at times $t_1, \ldots, t_{k_n}$ taking values in $\mathbb{R}^p$, whose definition can be extended to $[0,1]$ by linear interpolation.

Definition 9.9. *The multivariate standardized score process* $\{U_n^*(\beta(t), t),$ $t \in [0,1]\}$ *evaluated at* $\beta(t)$ *is such that, for* $j = 0, 1, \ldots, k_n$ *and* $t \in [t_j, t_{j+1}[$*,*

$$U_n^*(\beta(t), t) = U_n^*(\beta(t_j), t_j) + (tk_n - j)\left\{U_n^*(\beta(t_{j+1}), t_{j+1}) - U_n^*(\beta(t_j), t_j)\right\}.$$

Remark. We can readily construct several univariate processes on the basis of the multivariate model. Suppose for example that we have 2 binary covariates Z_1 and Z_2. If we stratify on the basis of Z_1 then we would have a single regression effect processes for Z_2 that can be studied, as in the simple case. The same is true if we stratify on the basis of Z_2 and study Z_1. If, instead of stratifying, we fit both covariables into the model, then, we can consider processes for each variable separately. We could also consider conditional processes, for example where we look at Z_2 and fix Z_1, leading to two conditional processes. And, of course, the prognostic index would define a joint process where both variables are included in a linear combination. Definition 9.9 is interesting because it allows us to take into account the degree of association between the covariates themselves as well as how they relate, individually and collectively, to survival outcome.

Working assumptions in the multivariate case

Recall that the norms of vectors in $\mathbb{R}^p$ and matrices in $\mathcal{M}_{p\times p}(\mathbb{R})$ are their sup norms, as defined in Appendix A. The function space $(D[0,1], \mathbb{R}^p)$ is equipped with the Skorokhod product topology. To establish the limit behavior of the process, we refer to Chauvel and O'Quigley (2014, 2017), and suppose the following conditions to hold:

B1 (Asymptotic stability). There exists $\delta_2>0$, a bounded neighborhood $\mathbb{B}'$ of β of radius δ_2 containing the zero function: $\mathbb{B}'=\left\{\gamma, \sup_{t\in[0,1]} \|\gamma(t)-\beta(t)\| < \delta_2\right\}$, and functions $s^{(r)}$ defined on $\mathbb{B}'\times[0,1]$ for $r=0,1,$ and 2, respectively, taking values in $\mathbb{R}$, $\mathbb{R}^p$, and the matrix space $\mathcal{M}_{p\times p}(\mathbb{R})$, such that

$$\sqrt{n} \sup_{t\in[0,1]\},\gamma\in\mathbb{B}'} \left\| S^{(r)}(\gamma(t),t) - s^{(r)}(\gamma(t),t) \right\| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

B2 (Asymptotic regularity). The deterministic functions $s^{(r)}$, $r=0,1,2$, defined in B1 are uniformly continuous for $t\in[0,1]$ and bounded in $\mathbb{B}'\times[0,1]$. Furthermore, for $r=0,1,2$ and $t\in[0,1]$, $s^{(r)}(\cdot,t)$ is a continuous function in $\mathbb{B}'$. The function $s^{(0)}$ is bounded below by a strictly positive constant.

B3 (Homoscedasticity). There exists a symmetric and positive definite matrix $\Sigma\in\mathcal{M}_{p\times p}(\mathbb{R})$ and a sequence of positive constants $(M_n)_n$ such that $\lim_{n\to\infty} M_n = 0$ which satisfy for all $\gamma\in\mathbb{B}'$, $t\in[0,1]$ and $i,j=1,\ldots,p$,

$$\left\| \frac{\partial^2}{\partial\beta_i\partial\beta_j} \mathcal{E}_\beta(Z\mid t)\Big|_{\beta=\gamma(t)} \right\| \le M_n \quad a.s., \tag{9.14}$$

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\},\gamma\in\mathbb{B}'} \left\| \mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2} - \Sigma^{1/2} \right\| \xrightarrow[n\to\infty]{P} 0. \tag{9.15}$$

B4 (Uniformly bounded covariates). There exists some $L'\in\mathbb{R}^{*+}$ such that

$$\sup_{i=1,\ldots,n} \sup_{u\in[0,\mathcal{T}]} \sup_{j=1,\ldots,p} \left| Z_i^{(j)}(u) \right| \le L'.$$

These conditions are extensions of hypotheses A1, A2, A3, and A4 from Section 9.4 to the multiple covariates case. We make the additional assumption that the conditional variance-covariance matrices do not depend on the parameter when it is close to the true regression parameter. This is a technical assumption required due to the use of the multidimensional Taylor-Lagrange inequality in the proofs. In the univariate case, we do not need this hypothesis thanks to the availability of a Taylor-Lagrange equality.

Proposition 9.4. (Chauvel 2014). *Under hypotheses B1, B3, and B4, we have the following:*

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\},\gamma\in\mathbb{B}'} \left\| \mathcal{V}_{\gamma(t)}(Z\mid t) - \Sigma \right\| \xrightarrow[n\to\infty]{P} 0. \tag{9.16}$$

The proposition is proved at the end of the chapter. The following theorem gives the limit behavior of the univariate regression effect process. Let $a=(a_1,\dots,a_p)$ be a function from $\mathbb{R}^p$ to $[0,1]$. Denote

$$\int_0^t a(s)ds=\left(\int_0^t a_1(s)ds,\dots,\int_0^t a_p(s)ds\right),\quad 0\le t\le 1.$$

Suppose $\gamma\in\mathbb{B}'$. Let $U_n^*(\gamma,\cdot)$ be the multivariate regression effect process calculated at γ. To construct this process, we consider $k_n=k_n(\gamma)$, which can be estimated by $\widehat{k_n}(\gamma)$. Recall that $k_n(\gamma)\le\widehat{k_n}(\gamma)$ almost surely (Equation 9.23) and by the definition of $\widehat{k_n}(\gamma)$ in the multivariate case, we have:

$$\forall t\in\{t_1,\dots,t_{k_n}\},\ \left\|\mathcal{V}_{\gamma(t)}(Z\mid t)^{-1}\right\|\le C_{\mathcal{V}}^{-1}\quad \text{a.s.} \tag{9.17}$$

Theorem 9.3. (Chauvel and O'Quigley 2014). *Let β_0 be a constant function on $\mathbb{B}'$. Under the non-proportional hazards model with regression coefficient $\beta(t)$, if the multivariate regression effect process is calculated at parameter β_0, we have the convergence in distribution result:*

$$U_n^*(\beta_0,\cdot)-\sqrt{k_n}B_n \quad \xrightarrow[n\to\infty]{\mathcal{L}} \quad \mathcal{W}_p(t), \tag{9.18}$$

where $\mathcal{W}_p(t)$ is a standard p-dimensional Brownian motion, and for each $t\in[0,1]$,

$$B_n(t)=\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\left\{\mathcal{E}_{\beta(t_i)}(Z\mid t_i)-\mathcal{E}_{\beta_0}(Z\mid t_i)\right\}.$$

Furthermore,

$$\sup_{t\in[0,1]}\left\|B_n(t)-\Sigma^{1/2}\int_0^t\{\beta(s)-\beta_0\}\,ds\right\| \quad \xrightarrow[n\to\infty]{P} \quad 0. \tag{9.19}$$

We recall that a p-dimensional Brownian motion is a vector of p independent Brownian motions, each in $\mathbb{R}$.

Remark. The constant $C_1(\beta,\beta_0)$ in Theorem 9.2 in which the convergence of the univariate process U_n^* is established has no multivariate equivalent. This is a consequence of Equation 9.14 in hypothesis B3, which is used in the proof in a multidimensional Taylor-Lagrange inequality, but is not needed in the univariate case. The constant C_2 in Theorem 9.2 is equivalent to the matrix $\Sigma^{1/2}$. In practice, for a large enough fixed value of n, the multivariate standardized score process $U_n^*(0,\cdot)$ can thus be approximated by a Brownian motion whose drift depends on the true parameter $\beta(t)$ of the model. Each component of the drift $\sqrt{k_n}\Sigma^{1/2}\int_0^t \beta(s)ds$ represents a linear combination of the cumulative effects $\int_0^t \beta_1(s)ds, \ldots, \int_0^t \beta_p(s)ds$, except when the variance-covariance matrix Σ is diagonal. Under the theorem's hypotheses, a direct consequence of Equation 9.16 is

$$\hat{\Sigma} := \frac{1}{k_n}\sum_{i=1}^{k_n} \mathcal{V}_{\beta_0}(Z \mid t_i) \quad \xrightarrow[n\to\infty]{P} \quad \Sigma, \tag{9.20}$$

which means that $\hat{\Sigma}$ is an estimator that converges to Σ. The following corollary allows us to decorrelate the components of the process U_n^* so that the drift of each component represents the corresponding effect (in $\beta_1(t), \ldots, \beta_p(t)$) separately.

Corollary 9.1. (Chauvel 2014). *Under the conditions of Theorem 9.3,*

$$\widehat{\Sigma}^{-1/2} U_n^*(\beta_0, \cdot) - \sqrt{k_n}\widehat{\Sigma}^{-1/2} B_n \quad \xrightarrow[n\to\infty]{\mathcal{L}} \quad \Sigma^{-1/2}\mathcal{W}_p,$$

and

$$\sup_{t\in[0,1]} \left\| \widehat{\Sigma}^{-1/2} B_n(t) - \int_0^t \{\beta(s) - \beta_0\}\, ds \right\| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

Thus, when there are several covariates, the drifts of the process $\widehat{\Sigma}^{-1/2} U_n^*(\beta_0, \cdot)$ help us to make inference on the regression coefficients. Each drift represents a cumulative coefficient and inference is done separately for each drift, as in the univariate case.

9.6 Iterated logarithm for effective sample size

This section is focused on a technical detail first studied by Chauvel (2014). The issue is thoroughly resolved if we are prepared to condition on the observed effective sample size. That would be our recommendation. If we do not wish to make use of such a simplifying argument, and we wish to appeal to results that are available in the framework of the martingale central limit theorem, then we need to pay some small amount of attention to a technical detail concerning effective sample size. The martingale central limit theorem requires us to make use of the concept of predictability. Our useable number of failure points, k_n, is, technically speaking, not predictable since, until the study is finished or until such a time as the variance becomes zero, we do not know what its value is. Our solution to this is simple, just take the value as fixed and known at its observed value. This amounts to conditioning on a future value, just like obtaining the Brownian bridge (not a martingale) from Brownian motion (a martingale) by conditioning on its final value. But, this very simple solution, while fine in our view, does take the inference away from under the umbrella of the martingale central limit theorem. A general solution is not hard to find but requires a little attention to some finicky details. We consider this more closely here. Let us suppose that there exists $\alpha_0 \in]0,1]$ such that

$$\frac{\widehat{k_n}(\beta)}{n} \xrightarrow[n\to\infty]{\text{a.s.}} \alpha_0.$$

Furthermore, suppose that there exists a sequence $(a_n)_n$ converging to 1 and $\widetilde{N} \in \mathbb{N}^*$ such that for each $n \geq \widetilde{N}$,

$$a_n \geq 1, \quad \frac{\widehat{k_n}(\beta)}{n} a_n \geq \alpha_0 \quad \text{a.s.} \tag{9.21}$$

This would imply in particular that $\widehat{k_n}(\beta)$ tends to infinity when n tends to infinity, i.e., the number of deaths increases without bound as the sample size increases. If Z is a discrete variable, this means that for a sample of infinite size, no group will run out of individuals. If Z is continuous, $\widehat{k_n}(\beta) = \sum_{i=1}^n \delta_i - 1$ or $\widehat{k_n}(\beta) = \sum_{i=1}^n \delta_i$ almost surely. If so, $n^{-1}\widehat{k_n}(\beta)$ is an estimator that converges to $P(T \leq C)$ and $\alpha_0 = P(T \leq C)$. The law of the iterated logarithm then allows us to construct a sequence (a_n) satisfying Equation 9.21.

Remark. Suppose we have one continuous covariate with censoring at the last observed time point, implying that $\widehat{k_n}(\beta) = \sum_{i=1}^n \delta_i = \sum_{i=1}^n \mathbf{1}_{X_i \leq C_i}$. The variables $\delta_1, \ldots, \delta_n$ are independent and identically distributed with finite second order moments. Their expectation is $\alpha_0 = P(T \leq C)$ and variance $\alpha_0(1-\alpha_0)$. Using the law of the iterated logarithm, we have:

$$\liminf_{n\to\infty} \frac{\sqrt{n}\left(n^{-1}\widehat{k_n}(\beta) - \alpha_0\right)}{\sqrt{2\alpha_0(1-\alpha_0)\log\log(n)}} = -1 \quad \text{a.s.}$$

Thus,

$$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N}^*,\ \forall n \geq N_\varepsilon, \quad \frac{\sqrt{n}\left(n^{-1}\widehat{k_n}(\beta) - \alpha_0\right)}{\sqrt{2\alpha_0(1-\alpha_0)\log\log(n)}} \geq -1-\varepsilon \quad \text{a.s.}$$

Setting $\varepsilon = 1/2$, $\exists N_{1/2} \in \mathbb{N}^*$, $\forall n \geq N_{1/2}$,

$$\frac{\sqrt{n}\left(n^{-1}\widehat{k_n}(\beta) - \alpha_0\right)}{\sqrt{2\alpha_0(1-\alpha_0)\log\log(n)}} \geq -\frac{3}{2} \quad \text{a.s.}, \qquad \sqrt{\frac{\log\log(n)}{n}} \leq \frac{1}{3}\sqrt{\frac{2\alpha_0}{1-\alpha_0}}. \tag{9.22}$$

Suppose that $n \geq N_{1/2}$. Then, with the help of Equation 9.22,

$$\frac{\widehat{k_n}(\beta)}{n} \geq \alpha_0 - \frac{3}{\sqrt{2n}}\sqrt{\alpha_0(1-\alpha_0)\log\log(n)},$$

and by setting

$$a_n = \left(1 - \frac{3}{\sqrt{2\alpha_0 n}}\sqrt{(1-\alpha_0)\log\log(n)}\right)^{-1},$$

we find $n^{-1}\widehat{k_n}(\beta)a_n \geq \alpha_0$. From Equation 9.22, $a_n \geq 1$ and we therefore have $\lim_{n\to\infty} a_n = 1$. We have thus constructed a sequence (a_n) which satisfies Equation 9.21.

The random variable $\widehat{k_n}(\beta)$ is known only at the end of data collection. It cannot be used during the experiment to construct a predictable process (in the sense of an increasing family of σ-algebras as described below). To overcome this we define the deterministic quantity $k_n(\beta) = n a_n^{-1}\alpha_0$, in order to obtain theoretical results. We have the almost sure convergence $\frac{\widehat{k_n}(\beta)}{k_n(\beta)} \underset{n\to\infty}{\longrightarrow} 1$, and Equation 9.21 implies that

$$\forall n \geq \widetilde{N}, \quad \widehat{k_n}(\beta) \geq k_n(\beta) \quad \text{a.s.} \tag{9.23}$$

9.7 Classwork and homework

1. Simulate 30 i.i.d. uniform variates, X_i, $i = 1,\ldots,30$. Calculate $S_i = \sum_{j=1}^{i} X_j$. Join, using straight lines, the points S_i and S_{i+1}. We call this a process. Make a graphical plot of the process based on S_i versus i, for $i = 1$ through $i = 30$. Evaluate theoretically, $E(S_{10})$, $E(S_{20})$, and $E(S_{30})$. Next evaluate Var(S_{10}), Var(S_{20}), and Var(S_{30}). Graph the theoretical quantities against their observed counterparts.

2. Repeat the experiment in the above question for $X_i^C = X_i - 0.5$.

3. Repeat the experiment of Question 1, for $X_i^C = (X_i - 0.5)/\sqrt{2.5}$. What can we say here? Where does the number 2.5 come from? If you had to make a guess at Var$(S_{19.5})$ what value would you choose?

4. Let $\mathcal{W}(t)$ be Brownian motion on the interval $(0,1)$. Consider the time points t where $t = 1/30, 2/30, \ldots, 29/30, 1$. Write down $E\mathcal{W}(t)$ and Var$\mathcal{W}(t)$ for these points. Compare these values with those calculated above.

5. Calculate $S_i^0 = S_i - i \times S_{30}/30$, based upon X_i^C. Make some general observations on this process.

6. Repeat each set of simulations; $X_i^C = (X_i - 0.5)/\sqrt{2.5}$, 1000 times. For each of these repetitions, note down the empirically observed averages of the means and variances at each value of i for $i = 1,\ldots,30$. Rescale the interval for i to be the interval (0,1). Plot the average mean and the average variance against $i/30$. What conclusions can be made?

7. Let X_i^C take the value $X_i/\sqrt{2.5}$ for i=1,...,10; the value $(X_i - 0.25)/\sqrt{2.5}$, for i=11,..., 20; and the value $(X_i - 0.50)/\sqrt{2.5}$ otherwise. Plot these series of values against i. What conclusions can be drawn? Calculate $S_i^0 = S_i - i \times S_{30}/30$. What can be said about this process?

8. The condition of homoscedasticity is helpful in obtaining several of the large sample approximations. In practice the condition is typically respected to a strong degree. Can you construct a situation in which the assumption may not be a plausible one? What would be the implications of this for inference?

9.8 Outline of proofs

We provide here some detail to the proofs needed to support the main findings of this chapter. Broad outlines for these proofs have been given in Chauvel and O'Quigley (2014, 2017) and further details can be found in Chauvel (2014).

Proposition 9.1 The counting process $\bar{N}$ on the initial time scale takes values in $\mathbb{N}$. Thus, for all $u \in [0, \mathcal{T}]$,

$$\{x \in \mathbb{R} : \bar{N}(u) \leq x\} = \{x \in \mathbb{R} : \bar{N}(u) \leq \lfloor x \rfloor\}.$$

The inverse process is such that $\bar{N}^{-1}(j)$ is the jth ordered time of death, $j = 1, \ldots, k_n$. Note that $\bar{N}(\bar{N}^{-1}(j)) = j$. Thus, by the definition of $\bar{N}^*$, we have for $0 \leq t \leq 1$,

$$\bar{N}^*(t) = \sum_{i=1}^{n} \mathbf{1}_{\{\bar{N}(X_i) \leq \lfloor k_n t \rfloor, \delta_i = 1\}} = \sum_{i=1}^{n} \mathbf{1}_{\{X_i \leq \bar{N}^{-1}(\lfloor k_n t \rfloor), \delta_i = 1\}}$$
$$= \bar{N}(\bar{N}^{-1}(\lfloor k_n t \rfloor)) = \lfloor k_n t$$

Lemma 9.1 Suppose that $\varepsilon > 0$. Note that

$$P\left\{ \sup_{0 \leq s \leq 1} \left| \frac{1}{k_n} \int_0^s A_n(t) d\bar{N}^*(t) - \int_0^s a(t)dt \right| > \varepsilon \right\}$$
$$\leq P\left\{ \sup_{0 \leq s \leq 1} \frac{1}{k_n} \left| \int_0^s \{A_n(t) - a(t)\} d\bar{N}^*(t) \right| > \frac{\varepsilon}{2} \right\}$$
$$+ P\left\{ \sup_{0 \leq s \leq 1} \left| \frac{1}{k_n} \int_0^s a(t) d\bar{N}^*(t) - \int_0^s a(t)dt \right| > \frac{\varepsilon}{2} \right\} \qquad (9.24)$$

Suppose also that $s \in [0, 1]$. For the first term we have:

$$\begin{aligned}
\frac{1}{k_n}\left|\int_0^s \{A_n(t)-a(t)\}\, d\bar{N}^*(t)\right| &= \frac{1}{k_n}\left|\sum_{i=1}^{\lfloor k_n s\rfloor}(A_n(t_i)-a(t_i))\right| \\
&\leq \frac{\lfloor k_n s\rfloor}{k_n}\sup_{i=1,\dots,\lfloor k_n s\rfloor}|A_n(t_i)-a(t_i)| \leq \sup_{i=1,\dots,k_n}|A_n(t_i)-a(t_i)|.
\end{aligned}$$

Therefore,

$$\sup_{0\leq s\leq 1}\frac{1}{k_n}\left|\int_0^s \{A_n(t)-a(t)\}\, d\bar{N}^*(t)\right| \leq \sup_{i=1,\dots,k_n}|A_n(t_i)-a(t_i)|,$$

and $\sup_{i=1,\dots,k_n}|A_n(t_i)-a(t_i)| \xrightarrow[n\to\infty]{P} 0$. As a result,

$$\lim_{n\to\infty} P\left\{\sup_{0\leq s\leq 1}\frac{1}{k_n}\left|\int_0^s \{A_n(t)-a(t)\}\, d\bar{N}^*(t)\right| > \frac{\varepsilon}{2}\right\} = 0.$$

For the second term on the right-hand side of Equation 9.24 we have:

$$\sup_{0\leq s\leq 1}\left|\frac{1}{k_n}\int_0^s a(t)d\bar{N}^*(t) - \int_0^s a(t)dt\right| = \sup_{0\leq s\leq 1}\left|\frac{1}{k_n}\sum_{i=1}^{\lfloor k_n s\rfloor} a(t_i) - \int_0^s a(t)dt\right|.$$

Note that a_n is piecewise constant on the interval $[0,1]$ such that $a_n(t) = a(t_i)$ for $t\in[t_i,t_{i+1}[$, $i=0,\dots,k_n$. We then have:

$$\frac{1}{k_n}\sum_{i=1}^{\lfloor k_n s\rfloor} a(t_i) = \int_0^s a_n(t)dt - \left(s-\frac{\lfloor k_n s\rfloor}{k_n}\right) a\left(\frac{\lfloor k_n s\rfloor}{k_n}\right).$$

As a result we have:

$$\begin{aligned}
&\sup_{0\leq s\leq 1}\left|\frac{1}{k_n}\int_0^s a(t)d\bar{N}^*(t) - \int_0^s a(t)dt\right| \\
&\leq \sup_{0\leq s\leq 1}\left|\int_0^s a_n(t)dt - \int_0^s a(t)dt\right| + \sup_{0\leq s\leq 1}\left|\left(s-\frac{\lfloor k_n s\rfloor}{k_n}\right) a\left(\frac{\lfloor k_n s\rfloor}{k_n}\right)\right| \\
&\leq \sup_{0\leq s\leq 1}\left|\int_0^s a_n(t)dt - \int_0^s a(t)dt\right| + \frac{1}{k_n}\sup_{0\leq s\leq 1}\left|a\left(\frac{\lfloor k_n s\rfloor}{k_n}\right)\right|,
\end{aligned}$$

Note that $\lim_{n\to\infty}\sup_{s\in[0,1]}|a_n(s)-a(s)| = 0$ by the uniform continuity of the function a. This implies uniform convergence

$$\lim_{n\to\infty}\sup_{s\in[0,1]}\left|\int_0^s a_n(t)dt - \int_0^s a(t)dt\right| = 0.$$

Furthermore, a is bounded so that $\sup_{0\le s\le 1}\left|a\left(\frac{\lfloor k_n s\rfloor}{k_n}\right)\right| < \infty$, which shows the result since

$$\lim_{n\to\infty}\sup_{0\le s\le 1}\left|\frac{1}{k_n}\int_0^s a(t)d\bar{N}^*(t) - \int_0^s a(t)dt\right| = 0, \tag{9.25}$$

Proposition 9.2 Let $t\in[0,1]$. The expectation of $\mathcal{Z}(t)$ (defined in (9.2)) conditional on the σ-algebra $\mathcal{F}^*_{t-}$ is

$$\begin{aligned}
E_{\beta(t)}(\mathcal{Z}(t)\mid\mathcal{F}^*_{t-}) &= E_{\beta(t)}\left(\sum_{i=1}^n Z_i(X_i)\mathbf{1}_{\phi_n(X_i)=t,\delta_i=1}\mid\mathcal{F}^*_{t-}\right)\\
&= E_{\beta(t)}\left(\sum_{i=1}^n Z_i\left(\phi_n^{-1}(t)\right)\mathbf{1}_{\phi_n(X_i)=t,\delta_i=1}\mid\mathcal{F}^*_{t-}\right)\\
&= \sum_{i=1}^n Z_i\left(\phi_n^{-1}(t)\right)E_{\beta(t)}\left(\mathbf{1}_{\phi_n(X_i)=t,\delta_i=1}\mid\mathcal{F}^*_{t-}\right)\\
&= \sum_{i=1}^n Z_i\left(\phi_n^{-1}(t)\right)\ P_{\beta(t)}\left(\phi_n(X_i)=t,\delta_i=1\mid\mathcal{F}^*_{t-}\right).
\end{aligned}$$

Indeed, Z and ϕ_n^{-1} are left continuous, so $Z_i\left(\phi_n^{-1}(t)\right) = Z_i\left(\phi_n^{-1}(t^-)\right)$ is $\mathcal{F}^*_{t-}$ measurable. By definition, the probability that individual i has time of death t under the model with parameter $\beta(t)$, conditional on the set of at-risk individuals at time t, is $P_{\beta(t)}\left(\phi_n(X_i)=t,\delta_i=1\mid\mathcal{F}^*_{t-}\right)=\pi_i(\beta(t),t)$. Thus,

$$E_{\beta(t)}(\mathcal{Z}(t)\mid\mathcal{F}^*_{t-}) = \mathcal{E}_{\beta(t)}(Z\mid t).$$

The same reasoning for the second-order moment gives the variance result.

Proposition 9.3 Let $i,j=1,2,\ldots,k_n,\ i<j$. By definition, the random variable

$$\mathcal{V}_{\beta(t_i)}(Z\mid t_i)^{-1/2}\left(\mathcal{Z}(t_i)-\mathcal{E}_{\beta(t_i)}(Z\mid t_i)\right)$$

is $\mathcal{F}^*_{t_i}$ measurable. As the σ-algebras satisfy $\mathcal{F}^*_{t_i}\subset\mathcal{F}^*_{t_j^-}$, this random variable is also $\mathcal{F}^*_{t_j^-}$ measurable. Furthermore, from Proposition 9.2,

$$\begin{aligned}
&E\left(\mathcal{V}_{\beta(t_j)}(Z\mid t_j)^{-1/2}\left(\mathcal{Z}(t_j)-\mathcal{E}_{\beta(t_j)}(Z\mid t_j)\right)\Big|\mathcal{F}^*_{t_j^-}\right)\\
&\quad=\mathcal{V}_{\beta(t_j)}(Z\mid t_j)^{-1/2}E\left(\mathcal{Z}(t_j)-\mathcal{E}_{\beta(t_j)}(Z\mid t_j)\Big|\mathcal{F}^*_{t_j^-}\right)=0.
\end{aligned}$$

Thus,

$$
\begin{aligned}
&E\left(\mathcal{V}_{\beta(t_i)}(Z\mid t_i)^{-1/2}\left\{\mathcal{Z}(t_i)-\mathcal{E}_{\beta(t_i)}(Z\mid t_i)\right\}\mathcal{V}_j^Z\left\{\mathcal{Z}(t_j)-\mathcal{E}_{\beta(t_j)}(Z\mid t_j)\right\}\right)\\
&=E\left[E\left(\mathcal{V}_{\beta(t_i)}(Z\mid t_i)^{-1/2}\left\{\mathcal{Z}(t_i)-\mathcal{E}_{\beta(t_i)}(Z\mid t_i)\right\}\mathcal{V}_j^Z\left\{\mathcal{Z}(t_j)-\mathcal{E}_{\beta(t_j)}(Z\mid t_j)\right\}\middle|\mathcal{F}^*_{t_j^-}\right)\right]\\
&=E\left[\mathcal{V}_{\beta(t_i)}(Z\mid t_i)^{-1/2}\left\{\mathcal{Z}(t_i)-\mathcal{E}_{\beta(t_i)}(Z\mid t_i)\right\}E\left(\mathcal{V}_j^Z\left(\mathcal{Z}(t_j)-\mathcal{E}_{\beta(t_j)}(Z\mid t_j)\right)\middle|\mathcal{F}^*_{t_j^-}\right)\right]=0.
\end{aligned}
$$

where we have written $\mathcal{V}_j^Z$ in place of $\mathcal{V}_{\beta(t_j)}(Z\mid t_j)^{-1/2}$

Lemma 9.2 Appealing to the law of large numbers, condition A1 implies that for $r=0,1,2$, we have:

$$
s^{(r)}(\gamma(0),0)=E\left(S^{(r)}(\gamma(0),0)\right)=E\left\{Y(0)Z(0)^r\exp(\gamma(0)Z(0))\right\},
$$

which is $E\{Z(0)^r\exp(\gamma(0)Z(0))\}$ since $Y(0)=\mathbf{1}_{X\geq 0}=1$. As a result,

$$
\begin{aligned}
v(\gamma(0),0)&=\frac{E\left\{Z(0)^2\exp(\gamma(0)Z(0))\right\}}{E\left\{\exp\left(\gamma(0)Z(0)\right)\right\}}-\left(\frac{E\left\{Z(0)\exp(\gamma(0)Z(0))\right\}}{E\left\{\exp\left(\gamma(0)Z(0)\right)\right\}}\right)^2\\
&=E\left\{Z(0)^2\Phi_{\gamma(0)}\left(Z(0)\right)\right\}-\left(E\left\{Z(0)\Phi_{\gamma(0)}\left(Z(0)\right)\right\}\right)^2,
\end{aligned}
$$

where

$$
\Phi_{\gamma(0)}\left(z\right)=\frac{\exp(\gamma(0)z)}{E\left(\exp\left(\gamma(0)Z(0)\right)\right)},\ z\in\mathbb{R}.
$$

Furthermore, denoting $P_{Z(0)}$ to be the distribution function of the random variable $Z(0)$ and X having distribution function P_X such that $dP_X(x)=\Phi_{\gamma(0)}(x)dP_{Z(0)}(x)$, $x\in\mathbb{R}$, we have:

$$
v(\gamma(0),0)=E(X^2)-E(X)^2=V(X).
$$

The function $\Phi_{\gamma(0)}$ being invertible on $\mathbb{R}$, X is distributed according to a Dirac law if and only if $Z(0)$ follows a Dirac law. We know that $V(Z(0))>0$ which is equivalent to saying that $Z(0)$ does not follow a Dirac law. Thus, X does not follow a Dirac law and $v(\gamma(0),0)=V(X)>0$. The hypothesis A3 concerning homoscedasticity indicates that, for all $t\in[0,1]$, $v(\gamma(t),t)=v(\gamma(0),0)$, which leads to the conclusion.

Lemma 9.3 Assume that conditions A1 and A2 hold and that $\gamma_1\in\mathbb{B}$. Recall the definition of $v(\gamma_1(\cdot),\cdot)$. We write $\sup_k^n$ to denote $\sqrt{n}\sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}}$ so that we have:

$$\begin{aligned}
&\sup_k^n \left|\mathcal{V}_{\gamma_1(t)}(Z \mid t) - v(\gamma_1(t),t)\right| \\
&= \sup_k^n \left| \frac{S^{(2)}(\gamma_1(t),t)}{S^{(0)}(\gamma_1(t),t)} - \left(\frac{S^{(1)}(\gamma_1(t),t)}{S^{(0)}(\gamma_1(t),t)}\right)^2 - \left[\frac{s^{(2)}(\gamma_1(t),t)}{s^{(0)}(\gamma_1(t),t)} - \left(\frac{s^{(1)}(\gamma_1(t),t)}{s^{(0)}(\gamma_1(t),t)}\right)^2\right]\right| \\
&\leq \sqrt{n} \sup_{t\in[0,1]} \left| \frac{S^{(2)}(\gamma_1(t),t)}{S^{(0)}(\gamma_1(t),t)} - \left(\frac{S^{(1)}(\gamma_1(t),t)}{S^{(0)}(\gamma_1(t),t)}\right)^2 - \left[\frac{s^{(2)}(\gamma_1(t),t)}{s^{(0)}(\gamma_1(t),t)} - \left(\frac{s^{(1)}(\gamma_1(t),t)}{s^{(0)}(\gamma_1(t),t)}\right)^2\right]\right| \\
&\leq V_n + W_n,
\end{aligned}$$

where

$$V_n = \sqrt{n} \sup_{t\in[0,1]} \left| \frac{S^{(2)}(\gamma_1(t),t)}{S^{(0)}(\gamma_1(t),t)} - \frac{s^{(2)}(\gamma_1(t),t)}{s^{(0)}(\gamma_1(t),t)} \right|,$$

and

$$W_n = \sqrt{n} \sup_{t\in[0,1]} \left| \left(\frac{S^{(1)}(\gamma_1(t),t)}{S^{(0)}(\gamma_1(t),t)}\right)^2 - \left(\frac{s^{(1)}(\gamma_1(t),t)}{s^{(0)}(\gamma_1(t),t)}\right)^2 \right|.$$

We can study the large sample behavior of these two terms separately. Denote m_0, M_1, and M_2 to be strictly positive constants such that $s^{(0)}(\gamma(t),t) \geq m_0$ et $\left|s^{(i)}(\gamma(t),t)\right| \leq M_i$, $(i=1,2)$ for all $t \in [0,1]$ and $\gamma \in \mathbb{B}$. Their existence follows from condition A2. We have:

$$\begin{aligned}
V_n &\leq \sqrt{n} \sup_{t\in[0,1]} \left| \frac{1}{S^{(0)}(\gamma_1(t),t)} \left(S^{(2)}(\gamma_1(t),t) - s^{(2)}(\gamma_1(t),t)\right)\right| \\
&\quad + \sqrt{n} \sup_{t\in[0,1]} \left| s^{(2)}(\gamma_1(t),t) \left(\frac{1}{S^{(0)}(\gamma_1(t),t)} - \frac{1}{s^{(0)}(\gamma_1(t),t)}\right)\right| \\
&\leq \sqrt{n} \sup_{t\in[0,1]} \frac{1}{S^{(0)}(\gamma_1(t),t)} \sup_{t\in[0,1]} \left|S^{(2)}(\gamma_1(t),t) - s^{(2)}(\gamma_1(t),t)\right| \\
&\quad + M_2\sqrt{n} \sup_{t\in[0,1]} \left| \frac{1}{S^{(0)}(\gamma_1(t),t)} - \frac{1}{s^{(0)}(\gamma_1(t),t)}\right|. \qquad (9.26)
\end{aligned}$$

Furthermore,

$$\begin{aligned}
&\sqrt{n} \sup_{t\in[0,1],\gamma\in\mathbb{B}} \left| \frac{1}{S^{(0)}(\gamma(t),t)} - \frac{1}{s^{(0)}(\gamma(t),t)} \right| \\
&\quad = \sqrt{n} \sup_{t\in[0,1],\gamma\in\mathbb{B}} \frac{\left|s^{(0)}(\gamma(t),t) - S^{(0)}(\gamma(t),t)\right|}{S^{(0)}(\gamma(t),t)\, s^{(0)}(\gamma(t),t)} \\
&\quad \leq m_0^{-1}\sqrt{n} \sup_{t\in[0,1],\gamma\in\mathbb{B}} \left|s^{(0)}(\gamma(t),t) - S^{(0)}(\gamma(t),t)\right| \sup_{t\in[0,1],\gamma\in\mathbb{B}} \frac{1}{S^{(0)}(\gamma(t),t)}.
\end{aligned}$$

From Equation 9.6, $\sqrt{n}\ \sup_{t\in[0,1],\gamma\in\mathbb{B}}\left|s^{(0)}(\gamma(t),t)-S^{(0)}(\gamma(t),t)\right|=o_P(1)$, and since $s^{(0)}$ is bounded below uniformly by m_0, we have $\sup_{t\in[0,1],\gamma\in\mathbb{B}} S^{(0)}(\gamma(t),t)^{-1}=O_P(1)$, which indicates that this term is bounded after a certain rank with high probability. As a result,

$$\sqrt{n}\sup_{t\in[0,1],\gamma\in\mathbb{B}}\left|\frac{1}{S^{(0)}(\gamma(t),t)}-\frac{1}{s^{(0)}(\gamma(t),t)}\right| \xrightarrow[n\to\infty]{P} 0. \tag{9.27}$$

In addition, under condition A1,

$$\sqrt{n}\sup_{t\in[0,1],\gamma\in\mathbb{B}}\left|S^{(2)}(\gamma(t),t)-s^{(2)}(\gamma(t),t)\right| \xrightarrow[n\to\infty]{P} 0,$$

and since $\sup_{t\in[0,1],\gamma\in\mathbb{B}} S^{(0)}(\gamma(t),t)^{-1}=O_P(1)$, we have convergence for

$$\sqrt{n}\sup_{t\in[0,1]}\frac{1}{S^{(0)}(\gamma_1(t),t)}\sup_{t\in[0,1]}\left|S^{(2)}(\gamma_1(t),t)-s^{(2)}(\gamma_1(t),t)\right| \xrightarrow[n\to\infty]{P} 0.$$

Equation 9.26 allows us to conclude that V_n converges in probability to 0 when $n\to\infty$. Similar arguments show that we can bound the term W_n by

$$\begin{aligned}
W_n &\le \sqrt{n}\sup_{t\in[0,1]}\left|\frac{1}{S^{(0)}(\gamma_1(t),t)^2}\left(S^{(1)}(\gamma_1(t),t)^2-s^{(1)}(\gamma_1(t),t)^2\right)\right|\\
&\quad+\sqrt{n}\sup_{t\in[0,1]}\left|s^{(1)}(\gamma_1(t),t)^2\left(\frac{1}{S^{(0)}(\gamma_1(t),t)^2}-\frac{1}{s^{(0)}(\gamma_1(t),t)^2}\right)\right|\\
&\le \sqrt{n}\sup_{t\in[0,1]}\frac{1}{S^{(0)}(\gamma_1(t),t)^2}\sup_{t\in[0,1]}\left|S^{(1)}(\gamma_1(t),t)^2-s^{(1)}(\gamma_1(t),t)^2\right|\\
&\quad+M_1^2\sqrt{n}\sup_{t\in[0,1]}\left|\frac{1}{S^{(0)}(\gamma_1(t),t)^2}-\frac{1}{s^{(0)}(\gamma_1(t),t)^2}\right|.
\end{aligned} \tag{9.28}$$

We have:

$$\begin{aligned}
&\sqrt{n}\sup_{t\in[0,1],\gamma\in\mathbb{B}}\left|\frac{1}{S^{(0)}(\gamma(t),t)^2}-\frac{1}{s^{(0)}(\gamma(t),t)^2}\right|\\
&\quad=\sqrt{n}\sup_{t\in[0,1],\gamma\in\mathbb{B}}\left|\left(\frac{1}{S^{(0)}(\gamma(t),t)}-\frac{1}{s^{(0)}(\gamma(t),t)}\right)\left(\frac{1}{S^{(0)}(\gamma(t),t)}+\frac{1}{s^{(0)}(\gamma(t),t)}\right)\right|\\
&\quad\le\sqrt{n}\sup_{t\in[0,1],\gamma\in\mathbb{B}}\left|\frac{1}{S^{(0)}(\gamma(t),t)}-\frac{1}{s^{(0)}(\gamma(t),t)}\right|\left(\sup_{t\in[0,1],\gamma\in\mathbb{B}}\frac{1}{S^{(0)}(\gamma(t),t)}+m_0^{-1}\right).
\end{aligned}$$

Equation 9.27 and the fact that $\sup_{t\in[0,1],\gamma\in\mathbb{B}} S^{(0)}(\gamma(t),t)^{-1}=O_P(1)$ imply that

$$\sqrt{n} \sup_{t\in[0,1],\gamma\in\mathbb{B}} \left| \frac{1}{S^{(0)}(\gamma(t),t)^2} - \frac{1}{s^{(0)}(\gamma(t),t)^2} \right| \xrightarrow[n\to\infty]{P} 0.$$

Similar arguments can be used to show that

$$\sqrt{n} \sup_{t\in[0,1],\gamma\in\mathbb{B}} \left| S^{(1)}(\gamma(t),t)^2 - s^{(1)}(\gamma(t),t)^2 \right| \xrightarrow[n\to\infty]{P} 0.$$

Since $\sup_{t\in[0,1],\gamma\in\mathbb{B}} S^{(0)}(\gamma(t),t)^{-2} = O_P(1)$, we have the convergence of

$$\sqrt{n} \sup_{t\in[0,1]} \frac{1}{S^{(0)}(\gamma_1(t),t)^2} \sup_{t\in[0,1]} \left| S^{(1)}(\gamma_1(t),t)^2 - s^{(1)}(\gamma_1(t),t)^2 \right| \xrightarrow[n\to\infty]{P} 0.$$

As a result, following Equation 9.28, W_n converges in probability to 0 when $n\to\infty$. In conclusion,

$$\sqrt{n} \sup_{t\in\{t_1,\dots,t_{k_n}\}} \left| \mathcal{V}_{\gamma_1(t)}(Z\mid t) - v(\gamma_1(t),t) \right| \xrightarrow[n\to\infty]{P} 0.$$

Under condition A3, there exists a positive constant $C(\gamma_1)$ such that $v(\gamma_1(t),t) = C(\gamma_1)$ for all $t\in[0,1]$. Lemma 9.2 indicates the strict positivity of $C(\gamma_1)$, and Equation 9.8 is thus demonstrated.

Equations 9.9 **and** 9.10. Let $\gamma_1\in\mathbb{B}$. Following Equation 9.8, under conditions A1, A2, and A3, there exist constants $C(\gamma_1),\ C(\beta_0) > 0$ such that

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\dots,t_{k_n}\}} \left| \mathcal{V}_{\gamma_1(t)}(Z\mid t) - C(\gamma_1) \right| \xrightarrow[n\to\infty]{P} 0, \tag{9.29}$$

$$\sqrt{n} \sup_{t\in\{t_1,t_2,\dots,t_{k_n}\}} \left| \mathcal{V}_{\beta_0}(Z\mid t) - C(\beta_0) \right| \xrightarrow[n\to\infty]{P} 0. \tag{9.30}$$

Since $k_n = k_n(\beta_0)$, Equation 9.9 is confirmed:

$$\forall t\in\{t_1,\dots,t_{k_n}\},\ \mathcal{V}_{\beta_0}(Z\mid t) > C_{\mathcal{V}} \quad \text{w.p. 1}$$

We then have:

$$\begin{aligned}
&\sup\nolimits_k^n \left| \frac{\mathcal{V}_{\gamma_1(t)}(Z\mid t)}{\mathcal{V}_{\beta_0}(Z\mid t)} - \frac{C(\gamma_1)}{C(\beta_0)} \right| \le \sup\nolimits_k^n \left| \frac{1}{\mathcal{V}_{\beta_0}(Z\mid t)} \left(\mathcal{V}_{\gamma_1(t)}(Z\mid t) - C(\gamma_1) \right) \right| \\
&+ C(\gamma_1) \sup\nolimits_k^n \left| \frac{1}{\mathcal{V}_{\beta_0}(Z\mid t)} - \frac{1}{C(\beta_0)} \right| \\
&\le C_{\mathcal{V}}^{-1} \sup\nolimits_k^n \left| \mathcal{V}_{\gamma_1(t)}(Z\mid t) - C(\gamma_1) \right| + C(\gamma_1) \sup\nolimits_k^n \left| \frac{1}{\mathcal{V}_{\beta_0}(Z\mid t)} - \frac{1}{C(\beta_0)} \right|.
\end{aligned}$$

Moreover,

$$\sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}}\left|\frac{1}{\mathcal{V}_{\beta_0}(Z\mid t)}-\frac{1}{C(\beta_0)}\right| = \sup_k^n \frac{\left|\mathcal{V}_{\beta_0}(Z\mid t)-C(\beta_0)\right|}{C(\beta_0)\mathcal{V}_{\beta_0}(Z\mid t)}$$
$$\leq C(\beta_0)^{-1}C_{\mathcal{V}}^{-1}\sup_k^n\left|\mathcal{V}_{\beta_0}(Z\mid t)-C(\beta_0)\right|.$$

The result 9.30 then implies:

$$\sqrt{n}\sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}}\left|\frac{1}{\mathcal{V}_{\beta_0}(Z\mid t)}-\frac{1}{C(\beta_0)}\right| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

This convergence together with that of Equation 9.29 lead to the conclusion that

$$\sqrt{n}\sup_{t\in\{t_1,t_2,\ldots,t_{k_n}\}}\left|\frac{\mathcal{V}_{\gamma_1(t)}(Z\mid t)}{\mathcal{V}_{\beta_0}(Z\mid t)}-\frac{C(\gamma_1)}{C(\beta_0)}\right| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

The same arguments can be used to prove Equation 9.10.

Theorem 9.1 This is a special case of the more general proof given for Theorem 9.2.

Theorem 9.2 Let $t\in[0,1]$. Under hypotheses A1, A2, A3, and A4, we suppose that $\beta(t)$ is the true value for the model and that the standarized score, U_n^*, is evaluated at the value $\beta_0\in\mathbb{B}$. Recall the equalities of Proposition 9.2:

$$\mathcal{E}_{\beta(t)}(Z\mid t)=E_{\beta(t)}\left(\mathcal{Z}(t)\left|\mathcal{F}_{t-}^*\right.\right), \qquad \mathcal{V}_{\beta_0}(Z\mid t)=V_{\beta_0}\left(\mathcal{Z}(t)\left|\mathcal{F}_{t-}^*\right.\right).$$

Define the process $\{U_n^{**}(\beta_0,t), t\in[0,1]\}$ by

$$U_n^{**}(\beta_0,t)=\int_0^t\frac{\mathcal{Z}(s)-E_{\beta_0}\left(\mathcal{Z}(s)\left|\mathcal{F}_{s-}^*\right.\right)}{\left(k_nV_{\beta_0}\left(\mathcal{Z}(s)\left|\mathcal{F}_{s-}^*\right.\right)\right)^{1/2}}\,d\bar{N}^*(s), \quad 0\leq t\leq 1. \tag{9.31}$$

This process is piecewise constant with a jump at each time of event. The outline of the proof is the following. Firstly, we will decompose $U_n^{**}(\beta_0,\cdot)$ into 2 processes belonging to $(D[0,1],\mathbb{R})$. We will show that the first process converges in distribution to a Brownian motion by appealing to a theorem for the convergence of differences of martingales. The convergence of the second process can be obtained by making use of Lemma 9.1. The large sample behavior of the process $U_n^*(\beta_0,\cdot)$ belonging to $(C[0,1],\mathbb{R})$, which is the linear interpolation of the variables $\{U_n^{**}(\beta_0,t_i), i=1,\ldots,k_n\}$, will be obtained by showing that the difference between U_n^* and U_n^{**} tends in probability to 0 when $n\to\infty$. A Taylor-Lagrange expansion of the conditional expectation $E_{\beta(t)}\{\mathcal{Z}(t)\mid\mathcal{F}_{t-}^*\}$ at the point $\beta(t)=\beta_0$ gives:

$$E_{\beta_0}\left(\mathcal{Z}(t)\,\middle|\,\mathcal{F}^*_{t^-}\right) = E_{\beta(t)}\left(\mathcal{Z}(t)\,\middle|\,\mathcal{F}^*_{t^-}\right) + (\beta_0 - \beta(t))\,\frac{\partial}{\partial\beta}E_{\beta}\left(\mathcal{Z}(t)\,\middle|\,\mathcal{F}^*_{t^-}\right)\Big|_{\beta=\tilde{\beta}(t)}, \tag{9.32}$$

where $\tilde{\beta}$ belongs to the ball centered at β having radius $\sup_{t\in[0,1]}|\beta(t)-\beta_0|$. Recall that

$$\frac{\partial}{\partial\beta}E_{\beta}\left(\mathcal{Z}(t)\,\middle|\,\mathcal{F}^*_{t^-}\right) = V_{\beta}\left(\mathcal{Z}(t)\,\middle|\,\mathcal{F}^*_{t^-}\right).$$

Therefore $U_n^{**}(\beta_0,t) = X_n(t) + \sqrt{k_n}A_n(t)$, where

$$A_n(t) = \frac{1}{k_n}\int_0^t \{\beta(s)-\beta_0\}\,\frac{V_{\tilde{\beta}(s)}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)^{1/2}}\,d\bar{N}^*(s),$$

$$X_n(t) = \frac{1}{k_n^{1/2}}\int_0^t \frac{\mathcal{Z}(s) - E_{\beta(s)}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)^{1/2}}\,d\bar{N}^*(s).$$

We need to consider the limiting behavior of A_n. We have:

$$\begin{aligned}
&\sup_{t\in[0,1]}\left|A_n(t) - C_2\int_0^t\{\beta(s)-\beta_0\}\,ds\right| \\
&\le \sup_{t\in[0,1]}|\beta(t)-\beta_0|\ \sup_{t\in[0,1]}\left|\frac{1}{k_n}\int_0^t \frac{V_{\tilde{\beta}(s)}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)^{1/2}}\,d\bar{N}^*(s) - C_2 t\right| \\
&\le \delta_1 \sup_{t\in[0,1]}\left|\frac{1}{k_n}\int_0^t \frac{V_{\tilde{\beta}(s)}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)^{1/2}}\,d\bar{N}^*(s) - C_2 t\right|,
\end{aligned} \tag{9.33}$$

since $\beta_0 \in \mathbb{B}$ with radius δ_1 (condition A1). Under conditions A1, A2, and A3, Equation 9.10 of Lemma 9.3 indicates the existence of a constant $C_2 > 0$ such that

$$\sup_{s\in\{t_1,t_2,\ldots,t_{k_n}\}}\left|\frac{V_{\tilde{\beta}(s)}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)^{1/2}} - C_2\right| \quad \underset{n\to\infty}{\overset{P}{\longrightarrow}} \quad 0.$$

Lemma 9.1 then implies:

$$\sup_{t\in[0,1]}\left|\frac{1}{k_n}\int_0^t \frac{V_{\tilde{\beta}(s)}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s^-}\right)^{1/2}}\,d\bar{N}^*(s) - C_2 t\right| \quad \underset{n\to\infty}{\overset{P}{\longrightarrow}} \quad 0.$$

Equation (9.33) now indicates that the convergence of Equation 9.13 is shown. We see that X_n converges in distribution to Brownian motion by making use of the functional central limit theorem for the differences of martingales developed by Helland (1982) and described in Theorem C.4. Let $j = 1,\ldots,k_n$. We write ξ_{j,k_n} as the jth increment of X_n :

$$\xi_{j,k_n} = \frac{\mathcal{Z}(t_j) - E_{\beta(t_j)}\left(\mathcal{Z}(t_j)\,\middle|\,\mathcal{F}^*_{t_j^-}\right)}{\left[k_n V_{\beta_0}\left(\mathcal{Z}(t_j)\,\middle|\,\mathcal{F}^*_{t_j^-}\right)\right]^{1/2}}.$$

The increment ξ_{j,k_n} is $\mathcal{F}^*_{t_j}$-measurable and $E_{\beta(t_j)}\left(\xi_{j,k_n}\,\middle|\,\mathcal{F}^*_{t_j^-}\right) = 0$. The process $\bar{N}^*$ contains a jump at each time of an event $t_j = j/k_n$ (Proposition 9.1), we then have:

$$X_n(t) = \int_0^{\lfloor tk_n \rfloor / k_n} \frac{\mathcal{Z}(s) - E_{\beta(s)}\left(\mathcal{Z}(s) \mid \mathcal{F}^*_{s-}\right)}{\left\{k_n V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s-}\right)\right\}^{1/2}} d\bar{N}^*(s) = \sum_{j=1}^{\lfloor tk_n \rfloor} \xi_{j,k_n},$$

where the function $t \longrightarrow \lfloor tk_n \rfloor$ is positive, increasing with integer values and right continuous. To check the assumption of Theorem C.4 we need to consider the expectation, the variance, and the Lyapunov condition: We begin with the expectation.

a. (Expectation). The inclusion of the σ-algebras $\mathcal{F}^*_{t_{j-1}} \subset \mathcal{F}^*_{t_j^-}$ leads to

$$E_{\beta(t_{j-1})}\left(\xi_{j,k_n}\,\middle|\,\mathcal{F}^*_{t_{j-1}}\right) = E_{\beta(t_{j-1})}\left(E_{\beta(t_j)}\left(\xi_{j,k_n}\,\middle|\,\mathcal{F}^*_{t_j^-}\right)\middle|\,\mathcal{F}^*_{t_{j-1}}\right) = 0. \tag{9.34}$$

b. (Variance). Under the hypotheses A1, A2, and A3, following Lemma 9.3, there exists a constant $C_1(\beta,\beta_0) > 0$ such that

$$\sup_{j=1,\ldots,k_n} \left| \frac{V_{\beta(t_j)}\left(\mathcal{Z}(t_j)\,\middle|\,\mathcal{F}^*_{t_j^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(t_j)\,\middle|\,\mathcal{F}^*_{t_j^-}\right)} - C_1(\beta,\beta_0)^2 \right| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

In particular, $C_1(\beta,\beta_0) = 1$ whenever $\beta(t) = \beta_0$. Let $s \in \{t_1, t_2, \ldots, t_{k_n}\}$. From Equation 9.9,

$$V_{\beta_0}\left(\mathcal{Z}(s)\,\middle|\,\mathcal{F}^*_{s-}\right) > C_{\mathcal{V}} \quad \text{a.s.}$$

We also have, according to the hypothesis A4, the upper bound,

$$\begin{aligned} |\mathcal{Z}(s)| &= \left| \sum_{i=1}^n Z_i(X_i) \mathbf{1}_{\phi_n(X_i) = s, \delta_i = 1} \right| \\ &\le \sup_{i=1,\ldots,n} |Z_i(X_i)| \sum_{i=1}^n \mathbf{1}_{\phi_n(X_i) = s, \delta_i = 1} \le L, \end{aligned} \tag{9.35}$$

since $s \in \{t_1, t_2, \dots, t_{k_n}\}$ donc $\sum_{i=1}^n \mathbf{1}_{\phi_n(X_i) = s, \delta_i = 1} = 1$. As a result,

$$V_{\beta(s)}\left(\mathcal{Z}(s) \,\middle|\, \mathcal{F}^*_{s^-}\right) \leq E_{\beta(s)}\left(\mathcal{Z}(s)^2 \,\middle|\, \mathcal{F}^*_{s^-}\right) \leq L^2.$$

In conclusion, we have, almost surely, the upper bound,

$$\sup_{j=1,\dots,k_n} \frac{V_{\beta(t_j)}\left(\mathcal{Z}(t_j) \,\middle|\, \mathcal{F}^*_{t_j^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(t_j) \,\middle|\, \mathcal{F}^*_{t_j^-}\right)} \leq \frac{L^2}{C_{\mathcal{V}}},$$

and an application of the dominated convergence theorem implies that

$$\sup_{j=1,\dots,k_n} \left| \frac{V_{\beta(t_j)}\left(\mathcal{Z}(t_j) \,\middle|\, \mathcal{F}^*_{t_j^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(t_j) \,\middle|\, \mathcal{F}^*_{t_j^-}\right)} - C_1(\beta, \beta_0)^2 \right| \quad \xrightarrow[n\to\infty]{L^1} \quad 0. \qquad (9.36)$$

Next, we have that

$$\sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(\xi^2_{j,k_n} \,\middle|\, \mathcal{F}^*_{t_{j-1}}\right) \quad \xrightarrow[n\to\infty]{L^1} \quad C_1(\beta, \beta_0)^2\, t.$$

and, to keep things uncluttered, we first write $W_0^{\mathcal{F}}(t_j) = \dfrac{V_{\beta(t_j)}(\mathcal{Z}(t_j) \mid \mathcal{F}^*_{t_j^-})}{V_{\beta_0}(\mathcal{Z}(t_j) \mid \mathcal{F}^*_{t_j^-})}$,

then,

$$\begin{aligned}
&\left| \sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(\xi^2_{j,k_n} \,\middle|\, \mathcal{F}^*_{t_{j-1}}\right) - C_1(\beta, \beta_0)^2 t \right| \\
&= \left| \sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(E_{\beta(t_j)}\left(\xi^2_{j,k_n} \,\middle|\, \mathcal{F}^*_{t_j^-}\right) \,\middle|\, \mathcal{F}^*_{t_{j-1}}\right) - C_1(\beta, \beta_0)^2 t \right| \\
&= \left| \frac{1}{k_n} \sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(W_0^{\mathcal{F}}(t_j) \,\middle|\, \mathcal{F}^*_{t_{j-1}}\right) - C_1(\beta, \beta_0) t \right| \\
&\leq \frac{1}{k_n} \sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(\left| W_0^{\mathcal{F}}(t_j) - C_1(\beta, \beta_0)^2 \right| \,\middle|\, \mathcal{F}^*_{t_{j-1}}\right) + C_1(\beta, \beta_0)^2 \left| \frac{\lfloor tk_n \rfloor}{k_n} - t \right|.
\end{aligned}$$

As a result of the above we now have:

$$
\begin{aligned}
&E\left(\left|\sum_{j=1}^{\lfloor tk_n\rfloor} E_{\beta(t_{j-1})}\left(\xi_{j,k_n}^2 \,\middle|\, \mathcal{F}_{t_{j-1}}^*\right) - C_1(\beta,\beta_0)^2 t\right|\right)\\
&\leq \frac{1}{k_n}\sum_{j=1}^{\lfloor tk_n\rfloor} E\left(\left|W_0^{\mathcal{F}}(t_j) - C_1(\beta,\beta_0)^2\right|\right) + C_1(\beta,\beta_0)^2\left|\frac{\lfloor tk_n\rfloor}{k_n} - t\right|\\
&\leq \frac{\lfloor tk_n\rfloor}{k_n}\sup_{j=1,\dots,\lfloor tk_n\rfloor} E\left(\left|W_0^{\mathcal{F}}(t_j) - C_1(\beta,\beta_0)^2\right|\right) + C_1(\beta,\beta_0)^2\left|\frac{\lfloor tk_n\rfloor}{k_n} - t\right|\\
&\leq \frac{\lfloor tk_n\rfloor}{k_n} E\left(\sup_{j=1,\dots,\lfloor tk_n\rfloor}\left|W_0^{\mathcal{F}}(t_j) - C_1(\beta,\beta_0)^2\right|\right) + C_1(\beta,\beta_0)^2\left|\frac{\lfloor tk_n\rfloor}{k_n} - t\right|\\
&\leq \frac{\lfloor tk_n\rfloor}{k_n} E\left(\sup_{j=1,\dots,k_n}\left|W_0^{\mathcal{F}}(t_j) - C_1(\beta,\beta_0)^2\right|\right) + C_1(\beta,\beta_0)^2\left|\frac{\lfloor tk_n\rfloor}{k_n} - t\right|.
\end{aligned}
$$

Equation 9.36 then leads to the conclusion that $\sum_{j=1}^{\lfloor tk_n\rfloor} E_\beta\left(\xi_{j,k_n}^2 \,\middle|\, \mathcal{F}_{t_{j-1}}^*\right)$ converges in mean and, as a result, in probability to $C_1(\beta,\beta_0)^2 t$ when $n \to \infty$.

c. We have the convergence in mean of $\sum_{j=1}^{\lfloor tk_n\rfloor} E_{\beta(t_{j-1})}\left(\xi_{j,k_n}^3 \mid \mathcal{F}_{t_{j-1}}^*\right)$ to 0 when $n \to \infty$. Furthermore,

$$
\begin{aligned}
E\left(\left|\sum_{j=1}^{\lfloor tk_n\rfloor} E_{\beta(t_{j-1})}\left(\xi_{j,k_n}^3 \mid \mathcal{F}_{t_{j-1}}^*\right)\right|\right) &\leq \sum_{j=1}^{\lfloor tk_n\rfloor} E\left(E_{\beta(t_{j-1})}\left(\left|\xi_{j,k_n}^3\right| \mid \mathcal{F}_{t_{j-1}}^*\right)\right)\\
&= \sum_{j=1}^{\lfloor tk_n\rfloor} E\left(\left|\xi_{j,k_n}^3\right|\right), \qquad (9.37)
\end{aligned}
$$

and keeping in mind that for $j = 1, \dots, k_n$, then

$$
\xi_{j,k_n} = \frac{\mathcal{Z}(t_j) - E_{\beta(t_j)}\left(\mathcal{Z}(t_j) \mid \mathcal{F}_{t_j^-}^*\right)}{\left\{k_n V_{\beta_0}\left(\mathcal{Z}(t_j) \,\middle|\, \mathcal{F}_{t_j^-}^*\right)\right\}^{1/2}}.
$$

Equation 9.35 implies that

$$
E\left(\left|\mathcal{Z}(t) - E_{\beta(t)}\left(\mathcal{Z}(t)|\mathcal{F}_{t-}^*\right)\right|^3\right) \leq 8L^3.
$$

Also, on the basis of Equation 9.9, $t \in \{t_1, t_2, \ldots, t_{k_n}\}$,$V_{\beta_0}\left(\mathcal{Z}(t)|\mathcal{F}^*_{t-}\right)$ is strictly greater than $C_{\mathcal{V}}$. As a result, Equation 9.37 becomes:

$$E\left(\left|\sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(\xi^3_{j,k_n} \mid \mathcal{F}^*_{t_{j-1}}\right)\right|\right)$$
$$\leq \sum_{j=1}^{k_n} \frac{1}{(k_n C_{\mathcal{V}})^{3/2}} E\left(\left|\mathcal{Z}(t_j) - E_{\beta(t_j)}\left(\mathcal{Z}(t_j)|\mathcal{F}^*_{t_j^-}\right)\right|^3\right) \leq \frac{8L^3}{k_n^{1/2} C_{\mathcal{V}}^{3/2}}, \tag{9.38}$$

and $\sum_{j=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{j-1})}\left(\xi^3_{j,k_n} \mid \mathcal{F}^*_{t_{j-1}}\right)$ converges in mean to 0 when $n \to \infty$ and, in consequence, in probability. The Lyapunov condition is verified.

We can now apply Theorem C.4 and conclude that

$$X_n \quad \underset{n\to\infty}{\overset{\mathcal{L}}{\longrightarrow}} \quad C_1(\beta, \beta_0)\mathcal{W},$$

where the constant $C_1(\beta, \beta_0)^2$ is the ratio of two asymptotic conditional variances v evaluated at β and β_0. When $\beta = \beta_0$, we have $C_1(\beta, \beta_0) = 1$. We have the relation

$$\begin{aligned} U^*_n(\beta_0, \cdot) - \sqrt{k_n} A_n &= U^*_n(\beta_0, \cdot) - U^{**}_n(\beta_0, \cdot) + U^{**}_n(\beta_0, \cdot) - \sqrt{k_n} A_n \\ &= U^*_n(\beta_0, \cdot) - U^{**}_n(\beta_0, \cdot) + X_n. \end{aligned} \tag{9.39}$$

We have then established the limiting behavior of X_n when $n \to \infty$. Next we look at the convergence in probability of $U^*_n(\beta_0, \cdot) - U^{**}_n(\beta_0, \cdot)$ to zero when $n \to \infty$ with respect to the uniform norm. Let $\varepsilon > 0$,

$$P\left(\|U^*_n(\beta_0, \cdot) - U^{**}_n(\beta_0, \cdot)\| \geq \varepsilon\right) = P\left(\sup_{t\in[0,1]} |U^*_n(\beta_0, t) - U^{**}_n(\beta_0, t)| \geq \varepsilon\right)$$
$$= P\left(\sup_{i=1,\ldots,k_n} |U^*_n(\beta_0, t_i) - U^{**}_n(\beta_0, t_{i-1})| \geq \varepsilon\right)$$
$$= P\left(\frac{1}{\sqrt{k_n}} \sup_{i=1,\ldots,k_n} \left|\frac{\mathcal{Z}(t_j) - E_{\beta_0}\left(\mathcal{Z}(t_j) \mid \mathcal{F}^*_{t_j^-}\right)}{V_{\beta_0}\left(\mathcal{Z}(t_j) \middle| \mathcal{F}^*_{t_j^-}\right)^{1/2}}\right| \geq \varepsilon\right) \leq P\left(\frac{2L}{\sqrt{k_n C_{\mathcal{V}}}} \geq \varepsilon\right),$$

Therefore, $U^*_n(\beta_0, \cdot) - U^{**}_n(\beta_0, \cdot) \underset{n\to\infty}{\overset{P}{\longrightarrow}} 0$. Slutsky's theorem applied to the decomposition of Equation 9.39 leads us to conclude that

$$U_n^*(\beta_0,\cdot)-{k_n}^{1/2}A_n \quad \underset{n\to\infty}{\overset{\mathcal{L}}{\longrightarrow}} \quad C_1(\beta,\beta_0)\mathcal{W},$$

in $(C[0,1],\mathbb{R})$ with respect to the topology of uniform convergence.

Proposition 9.4 Recall that $\|Ax\|\leq p\|A\|\|x\|$ et $\|AB\|\leq p\|A\|\|B\|$ for $A,B\in\mathcal{M}_{p\times p}(\mathbb{R})$ et $x\in\mathbb{R}^p$. Note that, if A is diagonal and its elements are positive, we have:

$$\left\|A^{1/2}\right\|=\max_{l=1,\dots,p}\left(A^{1/2}\right)_{ll}\leq\left(\max_{l=1,\dots,p}(A)_{ll}\right)^{1/2}=\|A\|^{1/2}.$$

Following the condition B3, Σ is a symmetric matrix having real coefficients so that $\Sigma^{1/2}$ exists and $\left\|\Sigma^{1/2}\right\|<\infty$. We have:

$$\begin{aligned}
&\sqrt{n}\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)-\Sigma\right\|\\
&=\sqrt{n}\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)-\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}\Sigma^{1/2}+\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}\Sigma^{1/2}-\Sigma\right\|\\
&\leq p\sqrt{n}\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}-\Sigma^{1/2}\right\|\\
&\times\left(\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}\right\|+\left\|\Sigma^{1/2}\right\|\right).
\end{aligned}$$

Now, for all $t\in\{t_1,\dots,t_{k_n}\}$ and $\gamma\in\mathbb{B}'$, according to the condition B4,

$$\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)\right\|\leq\left\|\mathcal{E}_\gamma\left(Z^{\otimes 2}\mid t\right)\right\|+\left\|\mathcal{E}_\gamma\left(Z\mid t\right)^{\otimes 2}\right\|\leq 2L'^2. \tag{9.40}$$

Therefore, in choosing an orthogonal transformation matrix $P_{\gamma(t)}(t)$ of norm 1 in the diagonalisation of the symmetric positive definite matrix $\mathcal{V}_{\gamma(t)}(Z\mid t)=P_{\gamma(t)}(t)\ D_{\gamma(t)}(t)\ P_{\gamma(t)}(t)^T$ with $D_{\gamma(t)}(t)$ diagonal, we have:

$$\begin{aligned}
&\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}\right\|\leq p^2\left\|P_{\gamma(t)}(t)\right\|^2\left\|D_{\gamma(t)}(t)^{1/2}\right\|\leq p^2\left\|D_{\gamma(t)}(t)\right\|^{1/2}\\
&=p^2\left\|P_{\gamma(t)}(t)^TP_{\gamma(t)}(t)D_{\gamma(t)}(t)P_{\gamma(t)}(t)^TP_{\gamma(t)}(t)\right\|^{1/2}\\
&\leq p^2\left(p^2\left\|P_{\gamma(t)}(t)D_{\gamma(t)}(t)P_{\gamma(t)}(t)^T\right\|\left\|P_{\gamma(t)}(t)\right\|^2\right)^{1/2}=p^3\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)\right\|^{1/2}\\
&\leq\sqrt{2}p^3L'.
\end{aligned}$$

from which, $\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}\right\|\leq\sqrt{2}p^3L'$, and

$$\begin{aligned}
&\sqrt{n}\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)-\Sigma\right\|\\
&\leq p\left(\sqrt{2}p^3L'+\left\|\Sigma^{1/2}\right\|\right)\sqrt{n}\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'}\left\|\mathcal{V}_{\gamma(t)}(Z\mid t)^{1/2}-\Sigma^{1/2}\right\|.
\end{aligned} \tag{9.41}$$

The result then follows from Equation 9.15.

Theorem 9.3 The proof follows the same outline as in the univariate setting. Recall that the space of cadlag functions of $(D[0,1],\mathbb{R}^p)$ is equipped with the product topology of Skorokhod, described in Definition A.5. Furthermore, the norm of the vector in $\mathbb{R}^p$ or a matrix of $\mathcal{M}_{p\times p}(\mathbb{R})$ is the sup norm. The process being evaluated at $\beta_0 \in \mathbb{B}'$, we have $k_n = k_n(\beta_0)$ and the upper bound of Equation 9.17 becomes:

$$\forall t \in \{t_1,\ldots,t_{k_n}\},\ \left\|\mathcal{V}_{\beta_0(t)}(Z \mid t)^{-1}\right\| \le C_{\mathcal{V}}^{-1} \quad \text{a.s.} \tag{9.42}$$

We take U_n^{**} to be a cadlag process having a jump at each event time $t_i = i/k_n, i=1,\ldots,k_n$, such that

$$U_n^{**}(\beta_0,t) = \frac{1}{\sqrt{k_n}} \sum_{i=1}^{\lfloor tk_n \rfloor} V_{\beta_0}\left(\mathcal{Z}(t_i)\,\middle|\,\mathcal{F}^*_{t_i-}\right)^{-1/2}\left\{\mathcal{Z}(t_i) - E_{\beta_0}\left(\mathcal{Z}(t_i)\,\middle|\,\mathcal{F}^*_{t_i-}\right)\right\}, 0 \le t \le 1.$$

Let $\gamma \in \mathbb{B}'$. Recall the equalities of Proposition 9.2:

$$\mathcal{E}_{\gamma(t_i)}(Z \mid t_i) = E_{\gamma(t_i)}\left(\mathcal{Z}(t_i)\,\middle|\,\mathcal{F}^*_{t_i-}\right), \qquad \mathcal{V}_{\gamma(t_i)}(Z \mid t_i) = V_{\gamma(t_i)}\left(\mathcal{Z}(t_i)\,\middle|\,\mathcal{F}^*_{t_i-}\right).$$

Note that $U_n^{**}(\beta_0,t) = X_n(t) + \sqrt{k_n}B_n(t)$ for $t \in [0,1]$,

$$X_n(t) = \frac{1}{\sqrt{k_n}} \sum_{i=1}^{\lfloor tk_n \rfloor} \mathcal{V}_{\beta_0}(Z \mid t_i)^{-1/2}\left\{\mathcal{Z}(t_i) - \mathcal{E}_{\beta(t_i)}(Z \mid t_i)\right\}$$

$$B_n(t) = \frac{1}{k_n} \sum_{i=1}^{\lfloor tk_n \rfloor} \mathcal{V}_{\beta_0}(Z \mid t_i)^{-1/2}\left\{\mathcal{E}_{\beta(t_i)}(Z \mid t_i) - \mathcal{E}_{\beta_0}(Z \mid t_i)\right\}.$$

We have:

$$\begin{aligned}
&\left\|U_n^*(\beta_0,\cdot) - \sqrt{k_n}B_n - \mathcal{W}_p\right\| \\
&\quad \le \|U_n^*(\beta_0,\cdot) - U_n^{**}(\beta_0,\cdot)\| + \left\|U_n^{**}(\beta_0,\cdot) - \sqrt{k_n}B_n - \mathcal{W}_p\right\| \\
&\quad = \|U_n^*(\beta_0,\cdot) - U_n^{**}(\beta_0,\cdot)\| + \|X_n - \mathcal{W}_p\|.
\end{aligned} \tag{9.43}$$

Let us consider the limiting behavior of the two terms on the right of the inequality. Considering just the first term and letting $\varepsilon > 0$, we have:

$$\begin{aligned}
&P\left(\sup_{t\in[0,1]}\|U_n^*(\beta_0,t)-U_n^{**}(\beta_0,t)\|\geq\varepsilon\right)\\
&\quad=P\left(\sup_{i=1,\dots,k_n}\left\|U_n^{**}\left(\beta_0,\frac{i}{k_n}\right)-U_n^{**}\left(\beta_0,\frac{i-1}{k_n}\right)\right\|\geq\varepsilon\right)\\
&\quad=P\left(\frac{1}{\sqrt{k_n}}\sup_{i=1,\dots,k_n}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\left\{\mathcal{Z}(t_i)-\mathcal{E}_{\beta_0}(Z\mid t_i)\right\}\right\|\geq\varepsilon\right)\\
&\quad\leq P\left(\frac{p}{\sqrt{k_n}}\sup_{i=1,\dots,k_n}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\right\|\left\|\mathcal{Z}(t_i)-\mathcal{E}_{\beta_0}(Z\mid t_i)\right\|\geq\varepsilon\right). \qquad (9.44)
\end{aligned}$$

We also have, for $i=1,\dots,k_n$,

$$\begin{aligned}
\left\|V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)^{-1/2}\right\| &= \left\|V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)^{-1}V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)^{1/2}\right\|\\
&\leq p\left\|V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)^{-1}\right\|\left\|V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)^{1/2}\right\|\\
&\leq C_{\mathcal{V}}^{-1}\sqrt{2}p^4L', \qquad (9.45)
\end{aligned}$$

where the last inequality is obtained from Equations 9.41 and 9.42. Also we have:

$$\left\|\mathcal{Z}(t_i)-E_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)\right\|\leq\|\mathcal{Z}(t_i)\|+\left\|E_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)\right\|\leq 2L'.$$

The inequality 9.44 implies that

$$P\left(\sup_{t\in[0,1]}\|U_n^*(\beta_0,t)-U_n^{**}(\beta_0,t)\|\geq\varepsilon\right)\leq P\left(\frac{2\sqrt{2}}{\sqrt{k_n}}p^5C_{\mathcal{V}}^{-1}(L')^2\geq\varepsilon\right), \qquad (9.46)$$

and $\lim_{n\to\infty}P\left(\sup_{t\in[0,1]}\|U_n^*(\beta_0,t)-U_n^{**}(\beta_0,t)\|\geq\varepsilon\right)=0$. This implies the convergence in probability of $U_n^*(\beta_0,\cdot)-U_n^{**}(\beta_0,\cdot)$ to 0 when $n\to\infty$ in the space $(D[0,1],\mathbb{R}^p)$ equipped with the Skorohod product topology as a consequence of the results of Section A.5. We now consider the second term. The convergence in distribution of X_n to $\mathcal{W}_p$ is obtained by an application of the multivariate functional central limit theorem C.5 described in Helland (1982). Before applying this theorem we need to check some working hypotheses. Let $i\in\{1,2,\dots,k_n\}$ and $t\in[0,1]$. Note that

$$\xi_{i,k_n}=(\xi^1_{i,k_n},\dots,\xi^p_{i,k_n})=\frac{1}{\sqrt{k_n}}V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)^{-1/2}\left\{\mathcal{Z}(t_i)-E_{\beta(t_i)}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i-}\right)\right\}$$

The ith increment in $\mathbb{R}^p$ of the process X_n. We have $X_n(t) = \sum_{i=1}^{\lfloor tk_n \rfloor} \xi_{i,k_n}$. Note that ξ_{i,k_n} is $\mathcal{F}^*_{t_i}$-measurable and that $E_{\beta(t_i)}\left(\xi_{i,k_n} \,\middle|\, \mathcal{F}^*_{t_i^-}\right) = 0$. Let $l, m \in \{1,\ldots,p\}$, $l \neq m$. We have e_l the lth vector of the canonical base of $\mathbb{R}^p$: all of the elements are zero with the exception of the lth which is 1. Also, $e_l^T e_m = 0$, $e_l^T e_l = 1$ and $\|e_l\| = \max_{i=1,\ldots,p}(e_l)_i = 1$. For $i = 1,\ldots,k_n$, we have:

$$\xi^l_{i,k_n} = e_l^T \xi_{i,k_n} = \xi^T_{i,k_n} e_l \in \mathbb{R}. \tag{9.47}$$

a.' (Table of martingale differences.) The inclusion of the σ-algebras $\mathcal{F}^*_{t_{i-1}} \subset \mathcal{F}^*_{t_i^-}$ and the centering of the increments implies that

$$E_{\beta(t_{i-1})}\left(\xi^l_{i,k_n} \,\middle|\, \mathcal{F}^*_{t_{i-1}}\right) = E_{\beta(t_{i-1})}\left(E_{\beta(t_i)}\left(\xi^l_{i,k_n} \,\middle|\, \mathcal{F}^*_{t_i^-}\right) \,\middle|\, \mathcal{F}^*_{t_{i-1}}\right) = 0.$$

b.' (Non-correlation.) Using Equation 9.47, we have:

$$\begin{aligned}
&E_{\beta(t_i)}\left(\xi^l_{i,k_n}\xi^m_{i,k_n} \,\middle|\, \mathcal{F}^*_{t_i^-}\right) = e_l^T E_{\beta(t_i)}\left(\xi_{i,k_n}\xi^T_{i,k_n} \,\middle|\, \mathcal{F}^*_{t_i^-}\right) e_m \\
&= \frac{1}{k_n} e_l^T V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2} V_{\beta(t_i)}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right) V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2} e_m \\
&= \frac{1}{k_n} e_l^T V\left(t_i^-, \beta_0, \beta(t_i)\right) e_m,
\end{aligned}$$

where

$$V\left(t_i^-, \beta_0, \beta(t_i)\right) = V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2} V_{\beta(t_i)}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right) V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2}.$$

Note that I_p is the identity matrix of $\mathcal{M}_{p\times p}(\mathbb{R})$. Also, $e_l^T I_p e_m = 0$. We have:

$$\begin{aligned}
&E\left(\left|\sum_{i=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{i-1})}\left(\xi^l_{i,k_n}\xi^m_{i,k_n} \,\middle|\, \mathcal{F}^*_{t_{i-1}}\right)\right|\right) \\
&= E\left(\left|\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n \rfloor} E_{\beta(t_{i-1})}\left(e_l^T V\left(t_i^-, \beta_0, \beta(t_i)\right) e_m \,\middle|\, \mathcal{F}^*_{t_{i-1}}\right)\right|\right) \\
&\leq \frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n \rfloor} E\left(\left|e_l^T V\left(t_i^-, \beta_0, \beta(t_i)\right) e_m\right|\right) \\
&= \frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n \rfloor} E\left(\left|e_l^T \left(V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p\right) e_m\right|\right)
\end{aligned}$$

$$
\begin{aligned}
&\leq \frac{\lfloor tk_n \rfloor}{k_n} \sup_{i=1,\ldots,\lfloor tk_n \rfloor} E\left(\left|e_l^T \left(V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right) e_m\right|\right) \\
&\leq \frac{\lfloor tk_n \rfloor}{k_n} E\left(\sup_{i=1,\ldots,\lfloor tk_n \rfloor} \left|e_l^T \left(V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right) e_m\right|\right) \\
&\leq p^2 \frac{\lfloor tk_n \rfloor}{k_n} \|e_l^T\| \|e_m\| E\left(\sup_{i=1,\ldots,\lfloor tk_n \rfloor} \left\|\left(V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right)\right\|\right) \\
&= p^2 \frac{\lfloor tk_n \rfloor}{k_n} E\left(\sup_{i=1,\ldots,\lfloor tk_n \rfloor} \left\|\left(V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right)\right\|\right) \\
&\leq p^2 \frac{\lfloor tk_n \rfloor}{k_n} E\left(\sup_{i=1,\ldots,k_n} \left\|\left(V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right)\right\|\right). \qquad (9.48)
\end{aligned}
$$

We need to show that

$$
\sup_{i=1,\ldots,k_n} \left\|V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right\| \quad \xrightarrow[n\to\infty]{L^1} \quad 0.
$$

We have:

$$
\begin{aligned}
&\sup_{i=1,\ldots,k_n} \left\|V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right\| \\
&= \sup_{i=1,\ldots,k_n} \left\| V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2} \left(V_{\beta(t_i)}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)\right.\right. \\
&\qquad \left.\left. - V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)\right) V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2} \right\| \\
&\leq p^2 \left(\sup_{i=1,\ldots,k_n} \left\| V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)^{-1/2}\right\|\right)^2 \sup_{i=1,\ldots,k_n} \left\| V_{\beta(t_i)}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right) \right. \\
&\qquad \left. - V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)\right\|.
\end{aligned}
$$

Equation 9.45 implies that

$$
\begin{aligned}
&\sup_{i=1,\ldots,k_n} \left\|V\left(t_i^-,\beta_0,\beta(t_i)\right) - I_p\right\| \\
&\leq 2p^{10} C_{\mathcal{V}}^{-2} \left(L'\right)^2 \sup_{i=1,\ldots,k_n} \left\| V_{\beta(t_i)}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right) - V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)\right\| \\
&\leq 2p^{10} C_{\mathcal{V}}^{-2} \left(L'\right)^2 \left(\sup_{i=1,\ldots,k_n} \left\| V_{\beta(t_i)}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right) - \Sigma\right\| + \right. \\
&\qquad \left. \sup_{i=1,\ldots,k_n} \left\| \Sigma - V_{\beta_0}\left(\mathcal{Z}(t_i) \,\middle|\, \mathcal{F}^*_{t_i^-}\right)\right\|\right).
\end{aligned}
$$

Finally, the convergence of Equation 9.16 implies the convergence in probability,

$$\sup_{i=1,\dots,k_n} \left\| V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

Equations 9.40 and 9.45 indicate that $\sup_{i=1,\dots,k_n} \left\| V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\|$ is bounded by a finite constant. The convergence is then again a convergence in mean.

$$\sup_{i=1,\dots,k_n} \left\| V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\| \quad \xrightarrow[n\to\infty]{L^1} \quad 0. \tag{9.49}$$

Equation 9.48 leads us to conclude that

$$\sum_{i=1}^{\lfloor k_n t \rfloor} E_{\beta(t_{i-1})}(\xi_{i,k_n}^l \xi_{i,k_n}^m | \mathcal{F}_{t_{i-1}^-}^*) \quad \xrightarrow[n\to\infty]{L^1} \quad 0,$$

and the convergence is one in probability.

c.' (Variance.) Similar arguments to a.' and b.' can be used, implying the equality,

$$\begin{aligned}
&\sum_{i=1}^{\lfloor t k_n \rfloor} E_{\beta(t_{i-1})}\left(\left(\xi_{i,k_n}^l\right)^2 \middle| \mathcal{F}_{t_{i-1}}^* \right) - t \\
&\quad = \frac{1}{k_n} \sum_{i=1}^{\lfloor t k_n \rfloor} E_{\beta(t_{i-1})}\left(e_l^T \left\{ V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\} e_l \middle| \mathcal{F}_{t_{i-1}}^* \right) + \left(\frac{\lfloor k_n t \rfloor}{k_n} - t \right).
\end{aligned}$$

This leads to

$$\begin{aligned}
&E\left(\left| \sum_{i=1}^{\lfloor t k_n \rfloor} E_{\beta(t_{i-1})}\left(\left(\xi_{i,k_n}^l\right)^2 \middle| \mathcal{F}_{t_{i-1}}^* \right) - t \right| \right) \\
&\quad \le \frac{1}{k_n} \sum_{i=1}^{\lfloor t k_n \rfloor} E\left(\left| e_l^T \left\{ V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\} e_l \right| \right) + \left| \frac{\lfloor k_n t \rfloor}{k_n} - t \right| \\
&\quad \le \frac{p^2}{k_n} \sum_{i=1}^{\lfloor t k_n \rfloor} E\left(\left\| V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\| \right) + \left| \frac{\lfloor k_n t \rfloor}{k_n} - t \right| \\
&\quad \le p^2 \frac{\lfloor k_n t \rfloor}{k_n} \sup_{i=1,\dots,k_n} E\left(\left\| V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\| \right) + \left| \frac{\lfloor k_n t \rfloor}{k_n} - t \right| \\
&\quad \le p^2 \frac{\lfloor k_n t \rfloor}{k_n} E\left(\sup_{i=1,\dots,k_n} \left\| V\left(t_i^-, \beta_0, \beta(t_i)\right) - I_p \right\| \right) + \left| \frac{\lfloor k_n t \rfloor}{k_n} - t \right|.
\end{aligned}$$

Equation 9.49 then leads to the conclusion that $\sum_{i=1}^{\lfloor t k_n \rfloor} E_{\beta(t_{i-1})}\left(\left(\xi_{i,k_n}^l\right)^2 \middle| \mathcal{F}_{t_{i-1}}^* \right) \quad \xrightarrow[n\to\infty]{L^1} \quad t$, indicating convergence in probability d.' (Lyapunov condition.) We have $\left|\xi_{i,k_n}^l\right| \leq \max_{l=1,\dots,p}\left|\xi_{i,k_n}^l\right| = \left\|\xi_{i,k_n}\right\|$. Therefore,

$$E\left(\sum_{i=1}^{\lfloor k_n t\rfloor} E_{\beta(t_{i-1})}\left(\left|\xi_{i,k_n}^l\right|^3 \middle| \mathcal{F}_{t_{i-1}}^*\right)\right) \leq \sum_{i=1}^{\lfloor k_n t\rfloor} E\left(\left|\xi_{i,k_n}^l\right|^3\right) \leq \sum_{i=1}^{k_n} E\left(\left\|\xi_{i,k_n}\right\|^3\right).$$

Using Equation 9.45 and condition B4, we have the bound,

$$\begin{aligned}
&E\left(\sum_{i=1}^{\lfloor k_n t\rfloor} E_{\beta(t_{i-1})}\left(\left|\xi_{i,k_n}^l\right|^3 \middle| \mathcal{F}_{t_{i-1}}^*\right)\right)\\
&\leq \frac{p^3}{k_n^{3/2}}\sum_{i=1}^{k_n} E\left(\left\|V_{\beta_0}\left(\mathcal{Z}(t_i)\,\middle|\,\mathcal{F}_{t_i-}^*\right)^{-1/2}\right\|^3 \left\|\mathcal{Z}(t_i) - E_{\beta(t_i)}\left(\mathcal{Z}(t_i)\,\middle|\,\mathcal{F}_{t_i-}^*\right)\right\|^3\right)\\
&\leq \frac{p^3}{k_n^{1/2}}\left(C_{\mathcal{V}}^{-1}\sqrt{2}p^4 L'\right)^3 (2L')^3.
\end{aligned}$$

In consequence, $\sum_{i=1}^{\lfloor k_n t\rfloor} E_{\beta(t_{i-1})}\left(\left|\xi_{i,k_n}^l\right|^3 \middle| \mathcal{F}_{t_{i-1}}^*\right)$ converge to 0 in mean and therefore in probability when $n \to \infty$.

In conclusion, all of the hypotheses of Theorem C.5 are met, which allows us to conclude that X_n converges to $\mathcal{W}_p$ when $n \to \infty$ in the space $(D[0,1],\mathbb{R}^p)$ equipped with the product topology of Skorohod. It then follows, as a result of Equation 9.43 and the convergence result of the first term on the right of Equation 9.46, Equation 9.18 is proved. To close the demonstration of the theorem it remains to prove Equation 9.19. Let $t \in [0,1]$. The multivariate Taylor-Lagrange inequality implies that for $s \in \{t_1,\dots,t_{k_n}\}$,

$$\left\|\mathcal{E}_{\beta(s)}(Z\mid s) - \mathcal{E}_{\beta_0}(Z\mid s) - \mathcal{V}_{\beta_0}(Z\mid s)\left(\beta(s)-\beta_0\right)\right\| \leq \frac{1}{2}M_n\left\|\beta(s)-\beta_0\right\|^2. \tag{9.50}$$

As a result,

$$\begin{aligned}
&\left\|B_n(t) - \Sigma^{1/2}\int_0^t \{\beta(s)-\beta_0\}\,ds\right\|\\
&= \left\|\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor} \mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\left\{\mathcal{E}_{\beta(t_i)}(Z\mid t_i) - \mathcal{E}_{\beta_0}(Z\mid t_i)\right\} - \Sigma^{1/2}\int_0^t \{\beta(s)-\beta_0\}\,ds\right\|\\
&\leq \frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\left\{\mathcal{E}_{\beta(t_i)}(Z\mid t_i) - \mathcal{E}_{\beta_0}(Z\mid t_i) - \mathcal{V}_{\beta_0}(Z\mid t_i)\{\beta(t_i)-\beta_0\}\right\}\right\|\\
&\quad + \frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\left\|\left(\mathcal{V}_{\beta_0}(Z\mid t_i)^{1/2} - \Sigma^{1/2}\right)\{\beta(t_i)-\beta_0\}\right\|
\end{aligned}$$

$$
\begin{aligned}
&+\left\|\Sigma^{1/2}\left(\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\beta(t_i)-\int_0^t\beta(s)ds+\beta_0\left(\frac{\lfloor tk_n\rfloor}{k_n}-t\right)\right)\right\|\\
&\le\frac{pM_n}{2k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\right\|\|\beta(t_i)-\beta_0\|^2+\frac{p}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{1/2}-\Sigma^{1/2}\right\|\|\beta(t_i)-\beta_0\|\\
&\quad+p\left\|\Sigma^{1/2}\right\|\left\|\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\beta(t_i)-\int_0^t\beta(s)ds\right\|+\left|\frac{\lfloor tk_n\rfloor}{k_n}-t\right|\left\|\Sigma^{1/2}\beta_0\right\|\\
&\le\frac{\lfloor tk_n\rfloor}{k_n}\frac{pM_n}{2}\sup_{i=1,\ldots,\lfloor tk_n\rfloor}\|\beta(t_i)-\beta_0\|^2\sup_{i=1,\ldots,\lfloor tk_n\rfloor}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{-1/2}\right\|\\
&\quad+p\frac{\lfloor tk_n\rfloor}{k_n}\sup_{i=1,\ldots,\lfloor tk_n\rfloor}\|\beta(t_i)-\beta_0\|\sup_{i=1,\ldots,\lfloor tk_n\rfloor}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{1/2}-\Sigma^{1/2}\right\|\\
&\quad+p\left\|\Sigma^{1/2}\right\|\sup_{l=1,\ldots,p}\left|\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\beta_l(t_i)-\int_0^t\beta_l(s)ds\right|+p\left|\frac{\lfloor tk_n\rfloor}{k_n}-t\right|\left\|\Sigma^{1/2}\right\|\|\beta_0\|.
\end{aligned}
$$

Since β_0 and $0\in\mathbb{B}'$ donc $\|\beta_0\|\le 2\delta_2$ et $\sup_{i=1,\ldots,\lfloor tk_n\rfloor}\|\beta(t_i)-\beta_0\|\le\sup_{t\in[0,1]}\|\beta(t)-\beta_0\|\le\delta_2$. Now, Equation 9.45 implies that

$$
\sup_{i=1,\ldots,k_n}\left\|V_{\beta_0}\left(\mathcal{Z}(t_i)\middle|\mathcal{F}^*_{t_i^-}\right)^{-1/2}\right\|\le\sqrt{2}C_{\mathcal{V}}^{-1}p^4L'.
$$

Therefore,

$$
\begin{aligned}
&\left\|B_n(t)-\Sigma^{1/2}\int_0^t\{\beta(s)-\beta_0\}\,ds\right\|\\
&\le M_n\frac{\lfloor tk_n\rfloor}{k_n}\frac{p^5}{\sqrt{2}}\delta_2^2C_{\mathcal{V}}^{-1}L'+p\frac{\lfloor tk_n\rfloor}{k_n}\delta_2\sup_{i=1,\ldots,k_n}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{1/2}-\Sigma^{1/2}\right\|\\
&\quad+p\left\|\Sigma^{1/2}\right\|\sup_{l=1,\ldots,p}\left|\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\beta_l(t_i)-\int_0^t\beta_l(s)ds\right|+2\delta_2p\left|\frac{\lfloor tk_n\rfloor}{k_n}-t\right|\left\|\Sigma^{1/2}\right\|.
\end{aligned}
$$

Bringing in the sup, we have:

$$
\begin{aligned}
&\sup_{t\in[0,1]}\left\|B_n(t)-\Sigma^{1/2}\int_0^t\{\beta(s)-\beta_0\}\,ds\right\|\\
&\le M_n\frac{p^5}{\sqrt{2}}\delta_2^2C_{\mathcal{V}}^{-1}L'+p\delta_2\sup_{i=1,\ldots,k_n}\left\|\mathcal{V}_{\beta_0}(Z\mid t_i)^{1/2}-\Sigma^{1/2}\right\|\\
&\quad+p\left\|\Sigma^{1/2}\right\|\sup_{t\in[0,1]}\sup_{l=1,\ldots,p}\left|\frac{1}{k_n}\sum_{i=1}^{\lfloor tk_n\rfloor}\beta_l(t_i)-\int_0^t\beta_l(s)ds\right|+\frac{2}{k_n}\delta_2p\left\|\Sigma^{1/2}\right\|.
\end{aligned}
\tag{9.51}
$$

According to B3, $\lim_{n\to\infty} M_n = 0$, and

$$\sup_{t\in\{t_1,t_2,\dots,t_{k_n}\},\gamma\in\mathbb{B}'} \left\| \mathcal{V}_{\gamma(t)}(Z \mid t)^{1/2} - \Sigma^{1/2} \right\| \quad \xrightarrow[n\to\infty]{P} \quad 0.$$

Let $l = 1,\dots,p$. The function β_l is bounded, allowing us to apply Equation 9.25 of the proof of Lemma 9.1 so that

$$\lim_{n\to\infty} \sup_{t\in[0,1]} \left| \frac{1}{k_n} \sum_{i=1}^{\lfloor tk_n \rfloor} \beta_l(t_i) - \int_0^t \beta_l(s)ds \right| = 0.$$

It follows that

$$\sup_{t\in[0,1]} \sup_{l=1,\dots,p} \left| \frac{1}{k_n} \sum_{i=1}^{\lfloor tk_n \rfloor} \beta_l(t_i) - \int_0^t \beta_l(s)ds \right| \leq \sum_{l=1}^{p} \sup_{t\in[0,1]} \left| \frac{1}{k_n} \sum_{i=1}^{\lfloor tk_n \rfloor} \beta_l(t_i) - \int_0^t \beta_l(s)ds \right|,$$

and since p is finite,

$$\lim_{n\to\infty} \sup_{t\in[0,1]} \sup_{l=1,\dots,p} \left| \frac{1}{k_n} \sum_{i=1}^{\lfloor tk_n \rfloor} \beta(t_i)_l - \int_0^t \beta(s)_l ds \right| = 0.$$

According to Equation 9.51, the convergence of Equation 9.19 is then established.